

Code ▾

Hassoun et al_ctDNA in Testicular Cancer Clinical Analysis

```
library(swimplot) library(coxphf) library(grid) library(gtable) library(readr) library(mosaic) library(dplyr) library(survival)
library(survminer) library(ggplot2) library(scales) library(coxphf) library(ggthemes) library(tidyverse)
library(gtsummary) library(flextable) library(parameters) library(car) library(ComplexHeatmap) library(tidyverse)
library(readxl) library(survival) library(janitor) library(rms) library(DT)
```

#ctDNA Detection rate by Stage and Window

Hide

```
#Pre-Orchiectomy
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$preSx.ctDNA.anytime!="",]
circ_data$preSx.ctDNA.anytime <- factor(circ_data$preSx.ctDNA.anytime, levels=c("NEGATIVE", "POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I", "II", "III"))
circ_data <- subset(circ_data, preSx.ctDNA.anytime %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$preSx.ctDNA.anytime == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$preSx.ctDNA.anytime, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$preSx.ctDNA.anytime == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

Stage <fctr>	Total_Count <int>	Positive_Count <int>	Rate <chr>
I	12	11	91.67%
II	1	1	100.00%
III	2	2	100.00%
Overall	15	14	93.33%

4 rows

Hide

```
#MRD Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.MRD.Window!="",]
circ_data$ctDNA.MRD.Window <- factor(circ_data$ctDNA.MRD.Window, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III"))
circ_data <- subset(circ_data, ctDNA.MRD.Window %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.MRD.Window == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.MRD.Window, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.MRD.Window == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

Stage <fctr>	Total_Count <int>	Positive_Count <int>	Rate <chr>
I	20	3	15.00%
II	9	7	77.78%
III	13	11	84.62%
Overall	42	21	50.00%

4 rows

Hide

```
#Surveillance Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.surveillance!="",]
circ_data$ctDNA.MRD.Window <- factor(circ_data$ctDNA.surveillance, levels=c("NEGATIVE", "POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I", "II", "III"))
circ_data <- subset(circ_data, ctDNA.surveillance %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.surveillance == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.surveillance, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.surveillance == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

Stage <fctr>	Total_Count <int>	Positive_Count <int>	Rate <chr>
I	27	4	14.81%
II	12	4	33.33%
III	12	5	41.67%
Overall	51	13	25.49%

4 rows

Hide

```

#Anytime post-surgery
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$postSx.ctDNA.anytime!="",]
circ_data$postSx.ctDNA.anytime <- factor(circ_data$postSx.ctDNA.anytime, levels=c("NEGATIVE", "POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I", "II", "III"))
circ_data <- subset(circ_data, postSx.ctDNA.anytime %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$postSx.ctDNA.anytime == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$postSx.ctDNA.anytime, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$postSx.ctDNA.anytime == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)

```

Stage<fctr>	Total_Count<int>	Positive_Count<int>	Rate<chr>
I	29	5	17.24%
II	17	12	70.59%
III	26	21	80.77%
Overall	72	38	52.78%
4 rows			

#Demographics Table

Hide

```

rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]

circ_data_subset <- circ_data %>%
  select(
    Age,
    pT,
    Stage,
    Histology,
    Treatment,
    RPLND.Pathology,
    preSx.markers.anytime,
    Markers.postSx.anytime,
    PFS.Event,
    OS.Event,
    OS.months) %>%
  mutate(
    Age = as.numeric(Age),
    pT = factor(pT, levels = c("0", "1", "2", "3", "X"), labels = c("pT0", "pT1", "pT2",
    "pT3", "pTX")),
    Stage = factor(Stage, levels = c("I", "II", "III")),
    Histology = factor(Histology, levels = c("Non seminoma", "Seminoma")),
    Treatment = factor(Treatment, levels = c("Surveillance", "Chemotherapy", "RPLND", "R
    PLND, Chemotherapy")),
    RPLND.Pathology = factor(RPLND.Pathology, levels = c("Negative", "Germ Cell/Teratoma")),
    preSx.markers.anytime = factor(preSx.markers.anytime, levels = c("NEGATIVE", "POSITIVE"), labels = c("Normal", "Elevated")),
    Markers.postSx.anytime = factor(Markers.postSx.anytime, levels = c("NEGATIVE", "POSITIVE"), labels = c("Normal", "Elevated")),
    PFS.Event = factor(PFS.Event, levels = c("FALSE", "TRUE"), labels = c("No Event", "Event")),
    OS.Event = factor(OS.Event, levels = c("FALSE", "TRUE"), labels = c("Alive", "Deceased")),
    OS.months = as.numeric(OS.months))
table1 <- circ_data_subset %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{median} ({p25} - {p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      Age ~ "Age",
      pT ~ "Pathological T Stage",
      Stage ~ "Overall Stage",
      Histology ~ "Histology",
      Treatment ~ "Post-surgery treatment",
      RPLND.Pathology ~ "RPLND Pathology",
      preSx.markers.anytime ~ "Serum Markers pre-op",
      Markers.postSx.anytime ~ "Serum Markers anytime post-op",
      PFS.Event ~ "Event During Follow-up",

```

```
OS.Event ~ "Vital Status",
OS.months ~ "Total Follow up (months)",) %>%
bold_labels()
table1
```

Characteristic	N = 74 ¹
Age	34 (27 - 39)
Pathological T Stage	
pT0	1 (1.4%)
pT1	39 (53%)
pT2	28 (38%)
pT3	3 (4.1%)
pTX	3 (4.1%)
Overall Stage	
I	30 (41%)
II	17 (23%)
III	27 (36%)
Histology	
Non seminoma	51 (69%)
Seminoma	23 (31%)
Post-surgery treatment	
Surveillance	17 (23%)
Chemotherapy	30 (41%)
RPLND	5 (6.8%)
RPLND, Chemotherapy	22 (30%)
RPLND Pathology	
Negative	10 (37%)
Germ Cell/Teratoma	17 (63%)
Unknown	47
¹ Median (Q1 - Q3); n (%)	

Characteristic	N = 74 ¹
Serum Markers pre-op	
Normal	19 (26%)
Elevated	55 (74%)
Serum Markers anytime post-op	
Normal	30 (41%)
Elevated	44 (59%)
Event During Follow-up	
No Event	39 (53%)
Event	35 (47%)
Vital Status	
Alive	69 (93%)
Deceased	5 (6.8%)
Total Follow up (months)	17 (12 - 25)
¹ Median (Q1 - Q3); n (%)	

Hide

```
fit1 <- as_flex_table(  
  table1,  
  include = everything(),  
  return_calls = FALSE  
)  
fit1
```

Characteristic	N = 74 ¹
Age	34 (27 - 39)
Pathological T Stage	
pT0	1 (1.4%)
pT1	39 (53%)
pT2	28 (38%)
pT3	3 (4.1%)
pTX	3 (4.1%)

¹Median (Q1 - Q3); n (%)

Characteristic	N = 74¹
Overall Stage	
I	30 (41%)
II	17 (23%)
III	27 (36%)
Histology	
Non seminoma	51 (69%)
Seminoma	23 (31%)
Post-surgery treatment	
Surveillance	17 (23%)
Chemotherapy	30 (41%)
RPLND	5 (6.8%)
RPLND, Chemotherapy	22 (30%)
RPLND Pathology	
Negative	10 (37%)
Germ Cell/Teratoma	17 (63%)
Unknown	47
Serum Markers pre-op	
Normal	19 (26%)
Elevated	55 (74%)
Serum Markers anytime post-op	
Normal	30 (41%)
Elevated	44 (59%)
Event During Follow-up	
No Event	39 (53%)
Event	35 (47%)
Vital Status	
Alive	69 (93%)
Deceased	5 (6.8%)
Total Follow up (months)	17 (12 - 25)

¹Median (Q1 - Q3); n (%)

[Hide](#)

```
save_as_docx(fit1, path= "~/Downloads/table1.docx")
```

#Heatmap with Clinical & Genomics Factors

[Hide](#)

```

rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data %>% arrange(Stage)
circ_datadf <- as.data.frame(circ_data)

ha <- HeatmapAnnotation(
  Stage = circ_data$Stage,
  Histology = circ_data$Histology,
  pT = circ_data$pT,
  Treatment = circ_data$Treatment,
  Platinum.Response = circ_data$Platinum.Response,
  PFS.Event = circ_data$PFS.Event,
  preSx.markers.anytime = circ_data$preSx.markers.anytime,
  Markers.MRD.Window = circ_data$Markers.MRD.Window,
  Markers.Surveillance = circ_data$Markers.Surveillance,
  Markers.post.ACT = circ_data$Markers.post.ACT,
  Markers.post.RPLND = circ_data$Markers.post.RPLND,
  Markers.postSx.anytime = circ_data$Markers.postSx.anytime,
  Markers.anytime = circ_data$Markers.anytime,
  preSx.ctDNA.anytime = circ_data$preSx.ctDNA.anytime,
  ctDNA.MRD.Window = circ_data$ctDNA.MRD.Window,
  ctDNA.surveillance = circ_data$ctDNA.surveillance,
  ctDNA.post.ACT = circ_data$ctDNA.post.ACT,
  ctDNA.post.RPLND = circ_data$ctDNA.post.RPLND,
  postSx.ctDNA.anytime = circ_data$postSx.ctDNA.anytime,
  ctDNA.anytime = circ_data$ctDNA.anytime,

  col = list(Stage = c("I" = "seagreen1", "II" = "orange", "III" = "purple"),
    Histology = c("Non seminoma" = "goldenrod", "Seminoma" = "blue4"),
    pT = c("0" = "coral", "1" = "coral", "2" = "darkgreen", "3" = "yellow3", "X" = "gray"),
    Treatment = c("Surveillance" = "lightblue", "Chemotherapy" = "orange2", "RPLND" = "khaki", "RPLND, Chemotherapy" = "brown2"),
    Platinum.Response = c("SD/PD" = "red3", "PR/CR" = "blue"),
    PFS.Event = c("TRUE" = "red3", "FALSE" = "blue"),
    preSx.markers.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.MRD.Window = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.Surveillance = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.post.ACT = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.post.RPLND = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.postSx.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    Markers.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    preSx.ctDNA.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.MRD.Window = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.surveillance = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.post.ACT = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.post.RPLND = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    postSx.ctDNA.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.anytime = c("POSITIVE" = "red3", "NEGATIVE" = "blue")
  )
)

```

```

)
ht <- Heatmap(matrix(nrow = 0, ncol = length(circ_data$Stage)), show_row_names = FALSE, cl
uster_rows = F, cluster_columns = FALSE, top_annotation = ha)
pdf("heatmap.pdf", width = 7, height = 7)
draw(ht, annotation_legend_side = "bottom")
dev.off()

```

```

null device
      1

```

#Overview plot

Hide

```

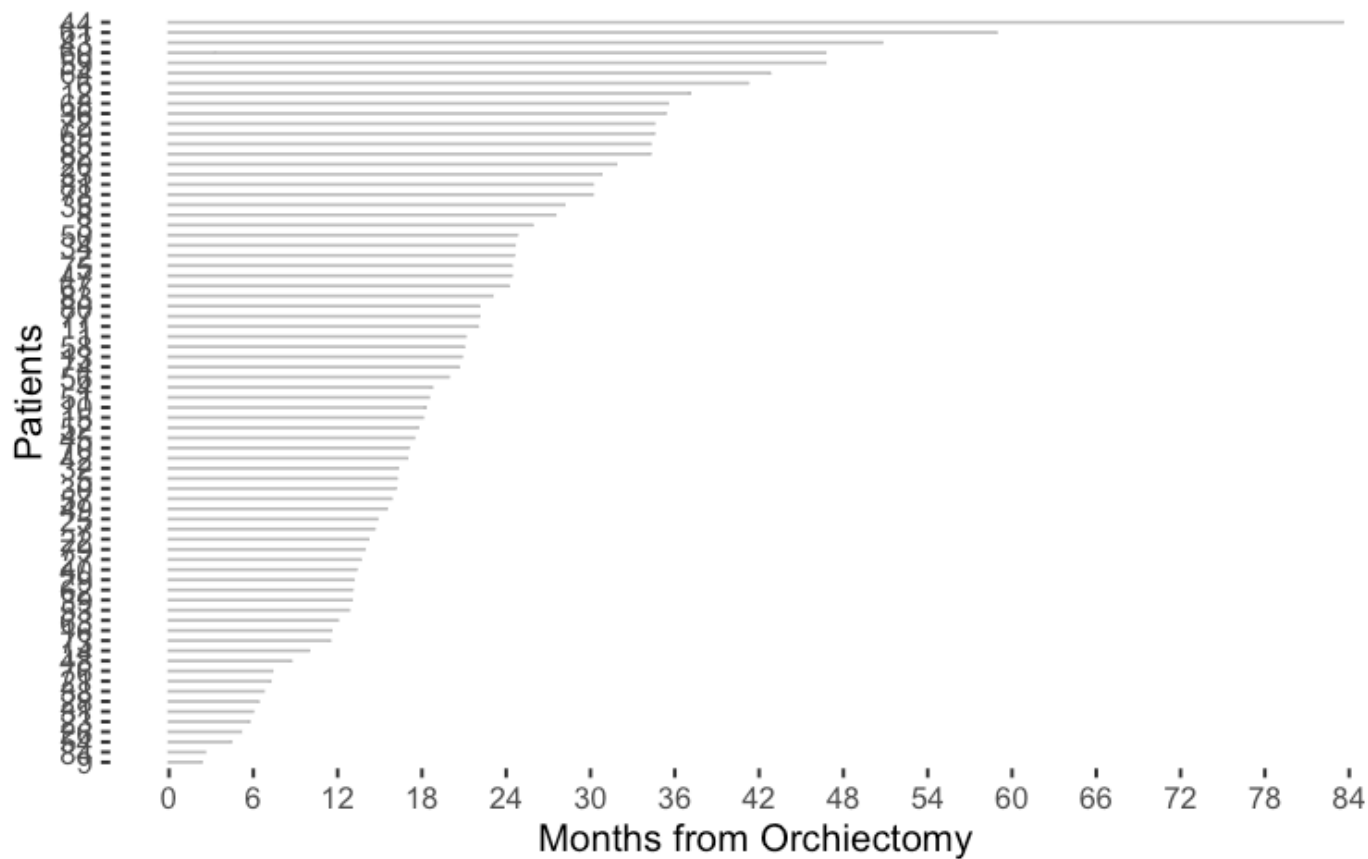
library(swimplot)
library(ggplot2)
library(grid)
library(gtable)

#Overview plot
rm(list=ls())
setwd("~/Downloads")
clinstage<- read.csv("CLIA Testicular_OP.csv")
clinstage_df<- as.data.frame(clinstage)

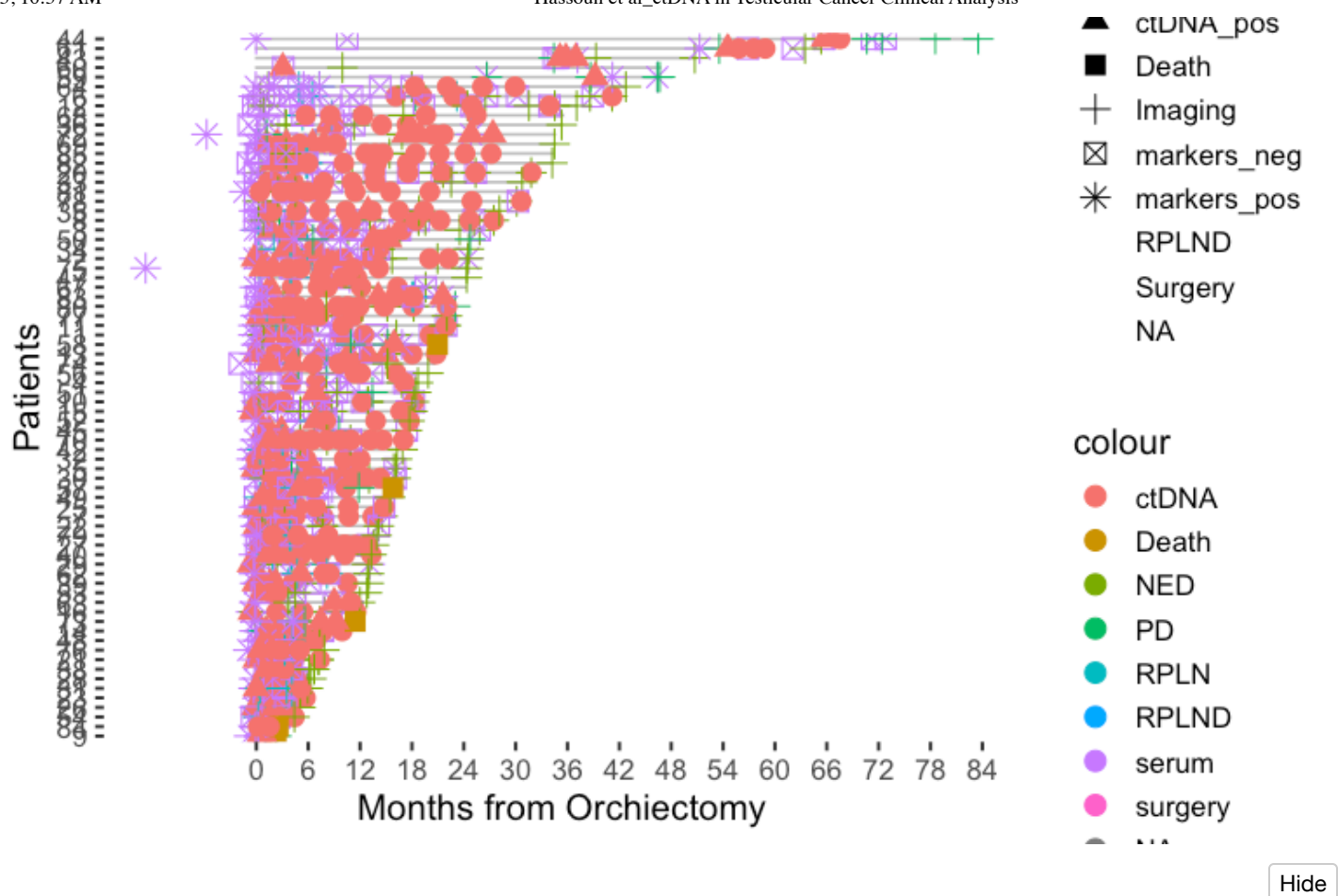
oplot_stratify <- swimmer_plot(df=clinstage_df,
                             id='PatientName',
                             end='fu.diff.months',
                             #name_fill='Arm',
                             col="gray",
                             alpha=0.75,
                             width=.01,
                             base_size = 14,
                             )

oplot_stratify <- oplot_stratify + theme(panel.border = element_blank())
oplot_stratify <- oplot_stratify + scale_y_continuous(breaks = seq(0, 108, by = 6))
oplot_stratify <- oplot_stratify + labs(x ="Patients" , y="Months from Orchiectomy")
oplot_stratify

```


[Hide](#)

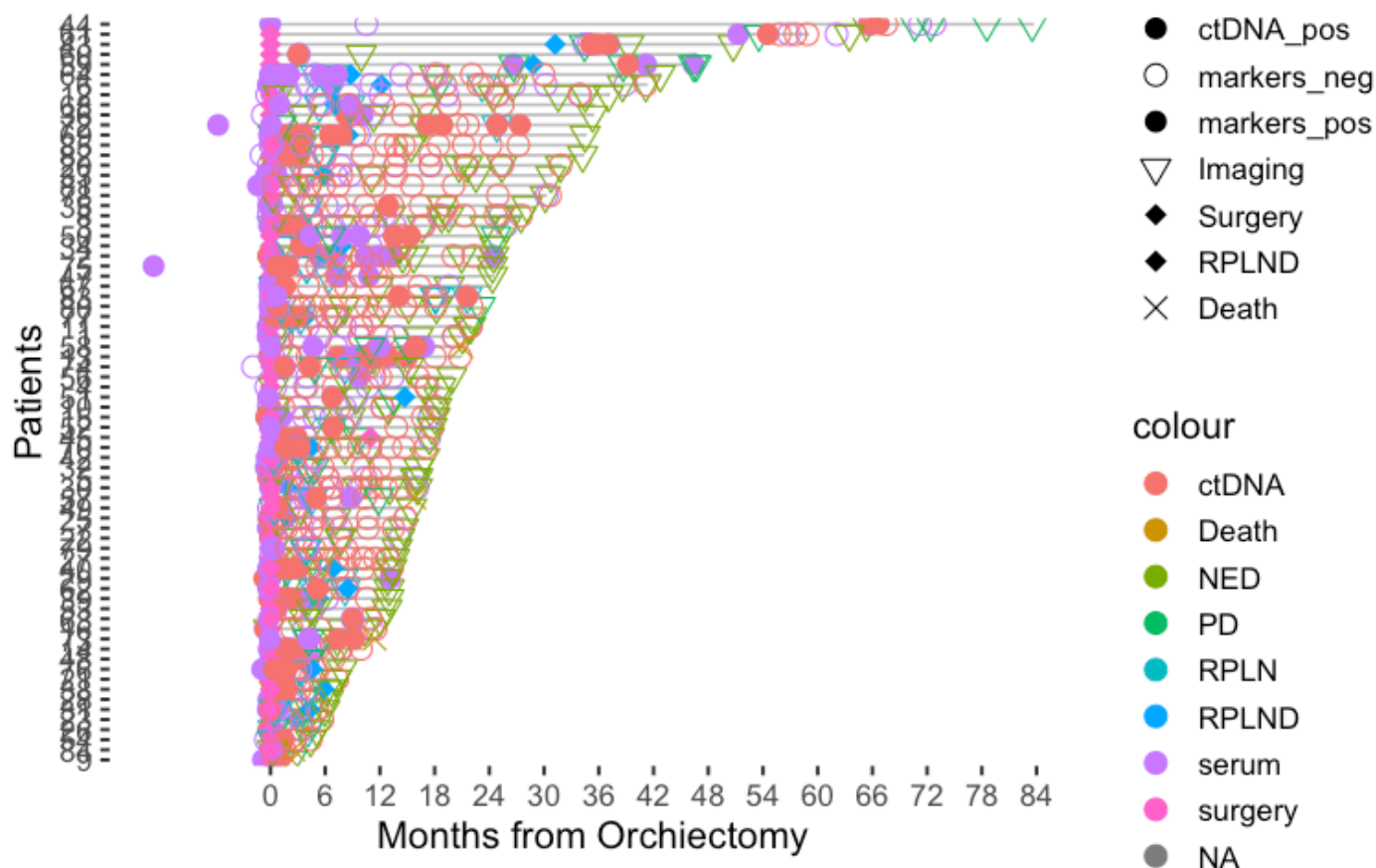
```
#plot events
oplot_ev3 <- oplot_stratify + swimmer_points(df_points=clinstage_df,
                                             id='PatientName',
                                             time='date.diff.months',
                                             name_shape = 'Event_type',
                                             name_col = 'Event',
                                             size=3.5,fill='black',
                                             #col='darkgreen'
)
oplot_ev3
```



```
#Shape customization to Event_type
```

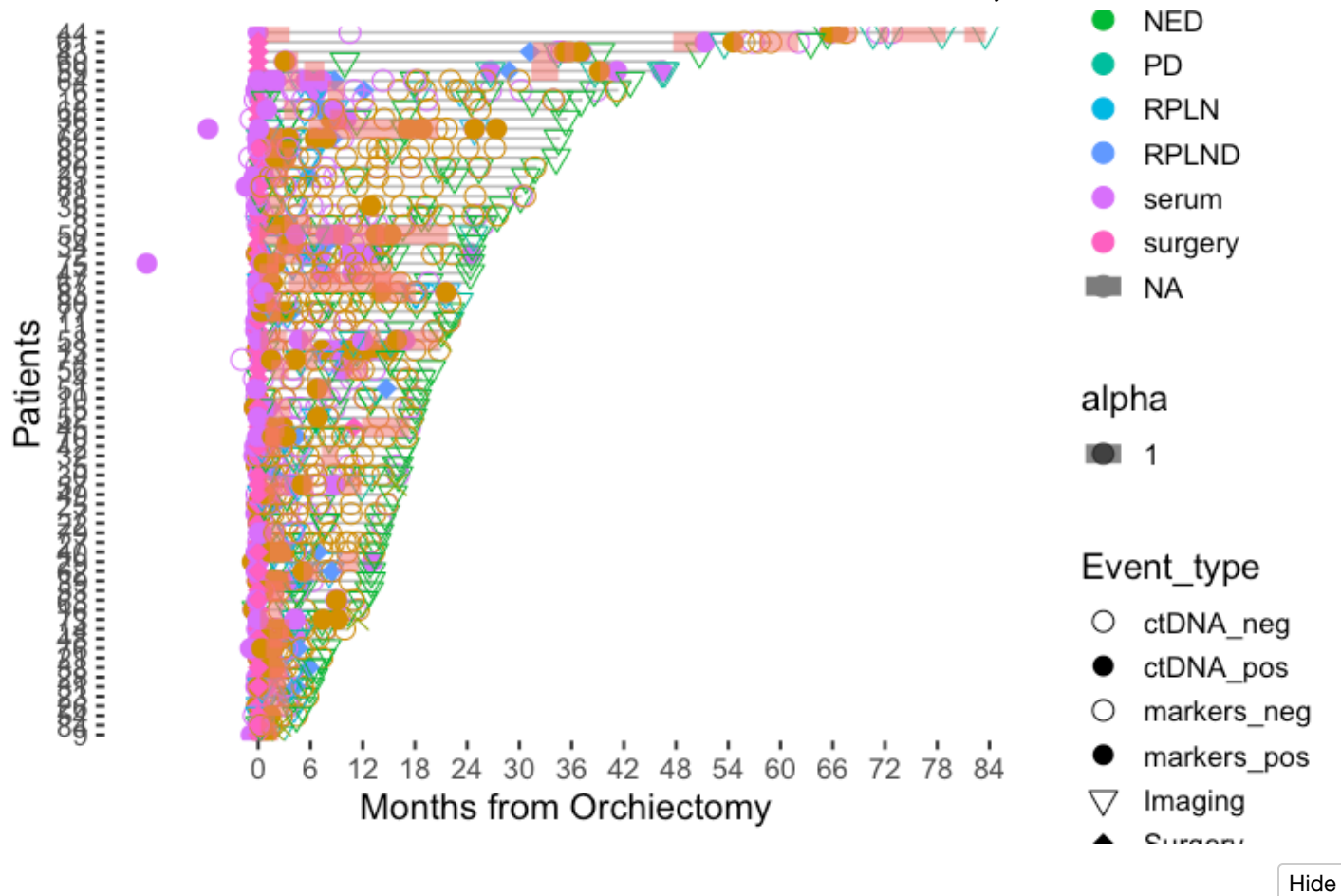
```
oplot_ev3.1 <- oplot_ev3 + ggplot2::scale_shape_manual(name="Event_type", values=c(1,16,
1,16,6,18,18,4), breaks=c('ctDNA_neg','ctDNA_pos','markers_neg','markers_pos','Imagin
g','Surgery','RPLND','Death'))
```

```
oplot_ev3.1
```

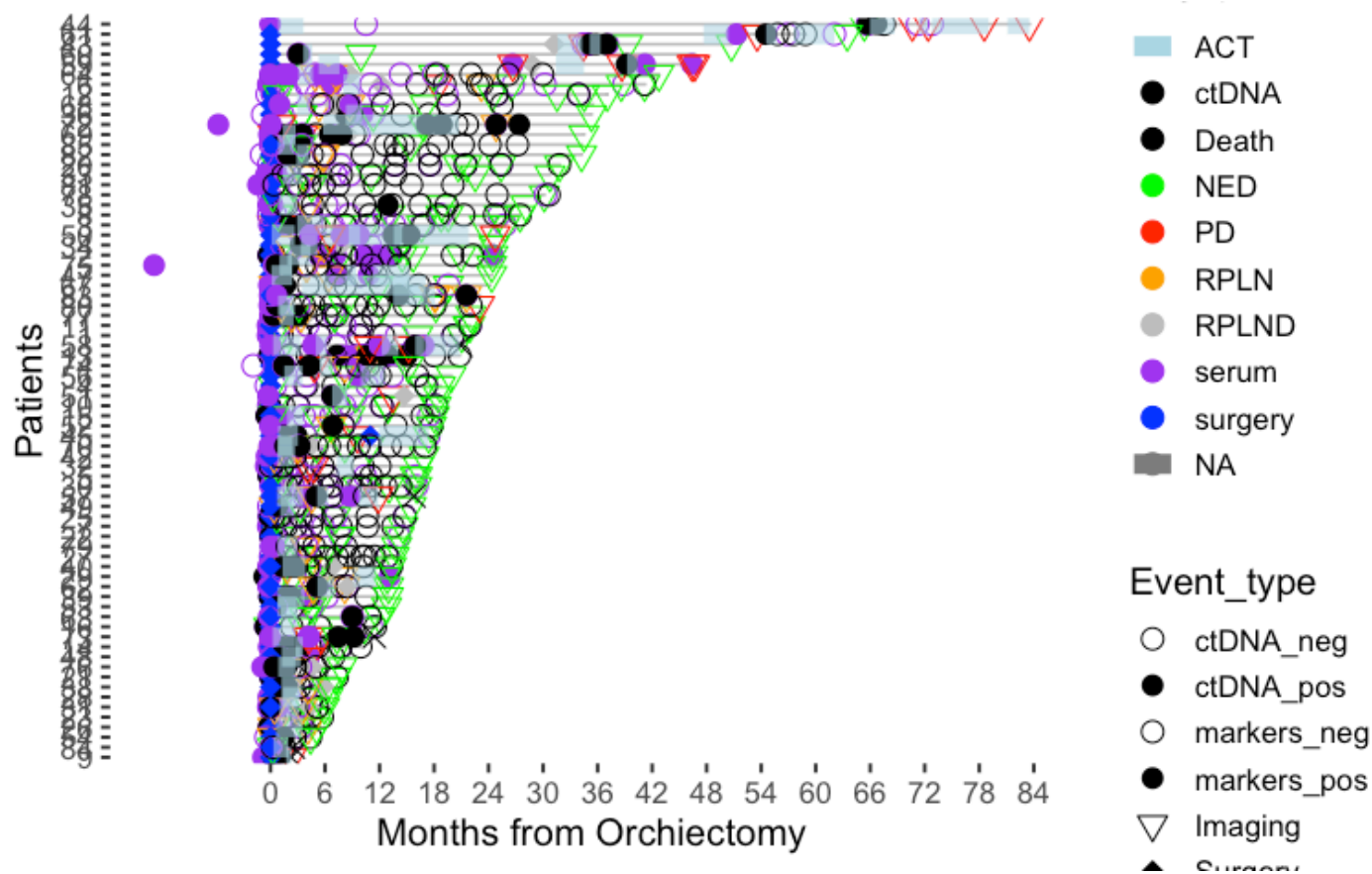

[Hide](#)

```
#plot treatment
oplot_ev4 <- oplot_ev3.1 + swimmer_lines(df_lines=clinstage_df,
                                         id='PatientName',
                                         start='Tx_start.months',
                                         end='Tx_end.months',
                                         name_col='Tx_type',
                                         size=3.5,
                                         name_alpha = 1.0)

oplot_ev4 <- oplot_ev4 + guides(linetype = guide_legend(override.aes = list(size = 5, color = "black")))
oplot_ev4
```



```
#colour customization
oplot_ev4.2 <- oplot_ev4 + ggplot2::scale_color_manual(name="Event",values=c( "lightblue", "black", "black", "green", "red", "orange", "grey", "purple", "blue"))
oplot_ev4.2
```

#Overview plot Stratified by Stage

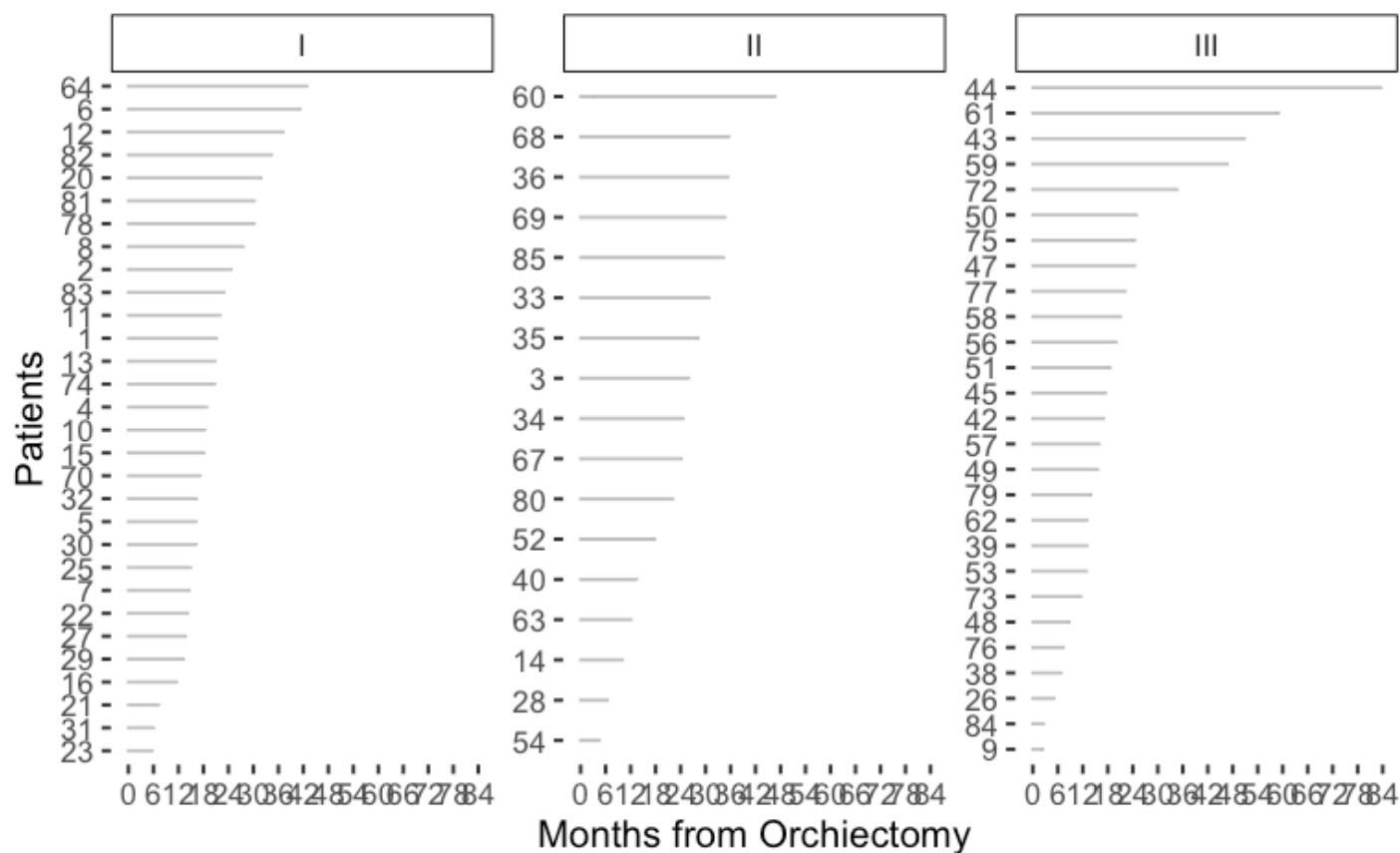
Hide

```
library(swimplot)
library(ggplot2)
library(grid)
library(gtable)

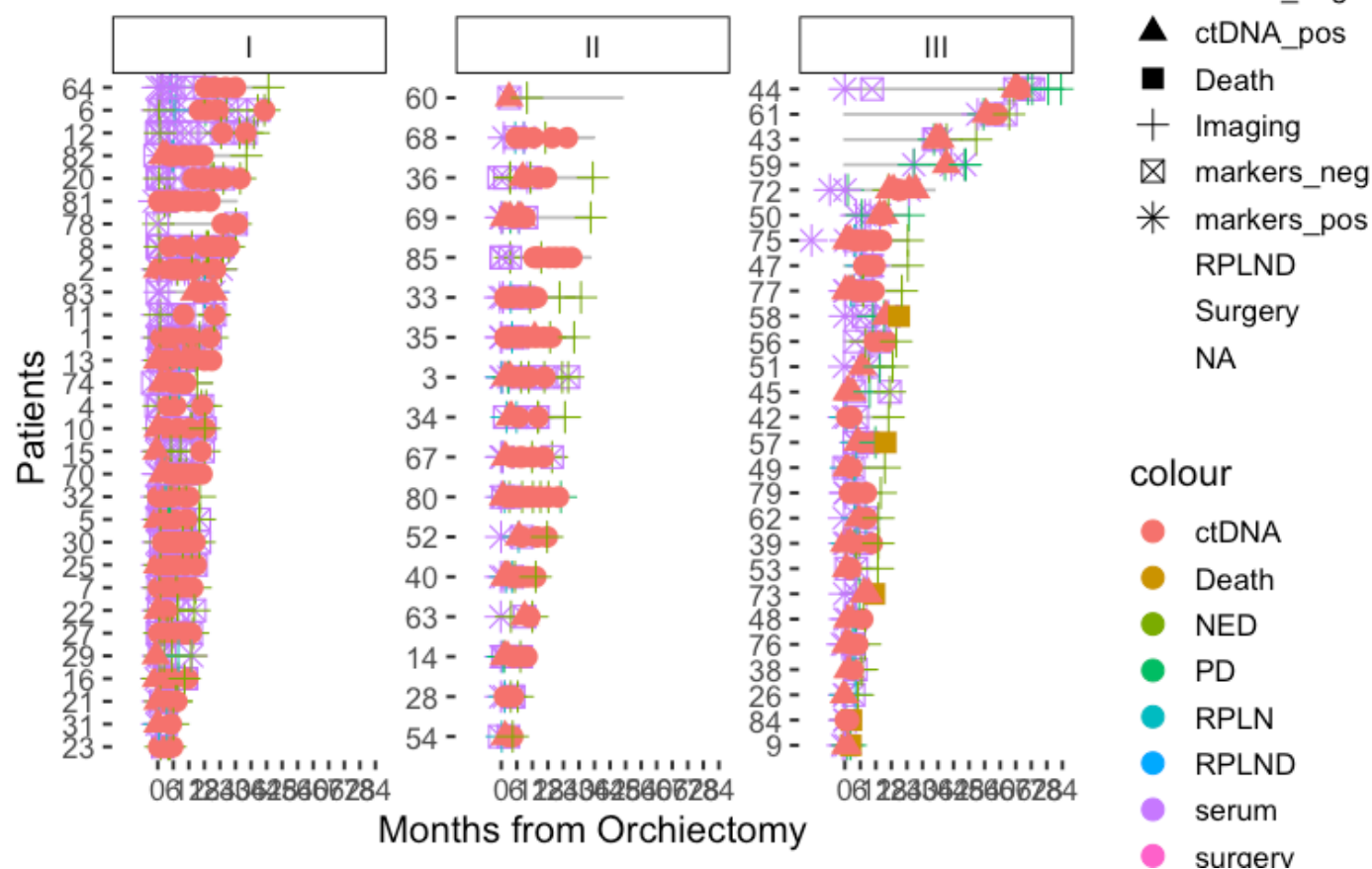
#Overview plot
rm(list=ls())
setwd("~/Downloads")
clinstage<- read.csv("CLIA Testicular_OP.csv")
clinstage_df<- as.data.frame(clinstage)

oplot_stratify <-swimmer_plot(df=clinstage_df,
                             id='PatientName',
                             end='fu.diff.months',
                             #name_fill='Arm',
                             col="gray",
                             alpha=0.75,
                             width=.01,
                             base_size = 14,
                             stratify= c('Stage'))

oplot_stratify <- oplot_stratify + theme(panel.border = element_blank())
oplot_stratify <- oplot_stratify + scale_y_continuous(breaks = seq(0, 108, by = 6))
oplot_stratify <- oplot_stratify + labs(x ="Patients" , y="Months from Orchiectomy")
oplot_stratify
```


[Hide](#)

```
#plot events
oplot_ev3 <- oplot_stratify + swimmer_points(df_points=clinstage_df,
                                             id='PatientName',
                                             time='date.diff.months',
                                             name_shape = 'Event_type',
                                             name_col = 'Event',
                                             size=3.5,fill='black',
                                             #col='darkgreen'
)
oplot_ev3
```

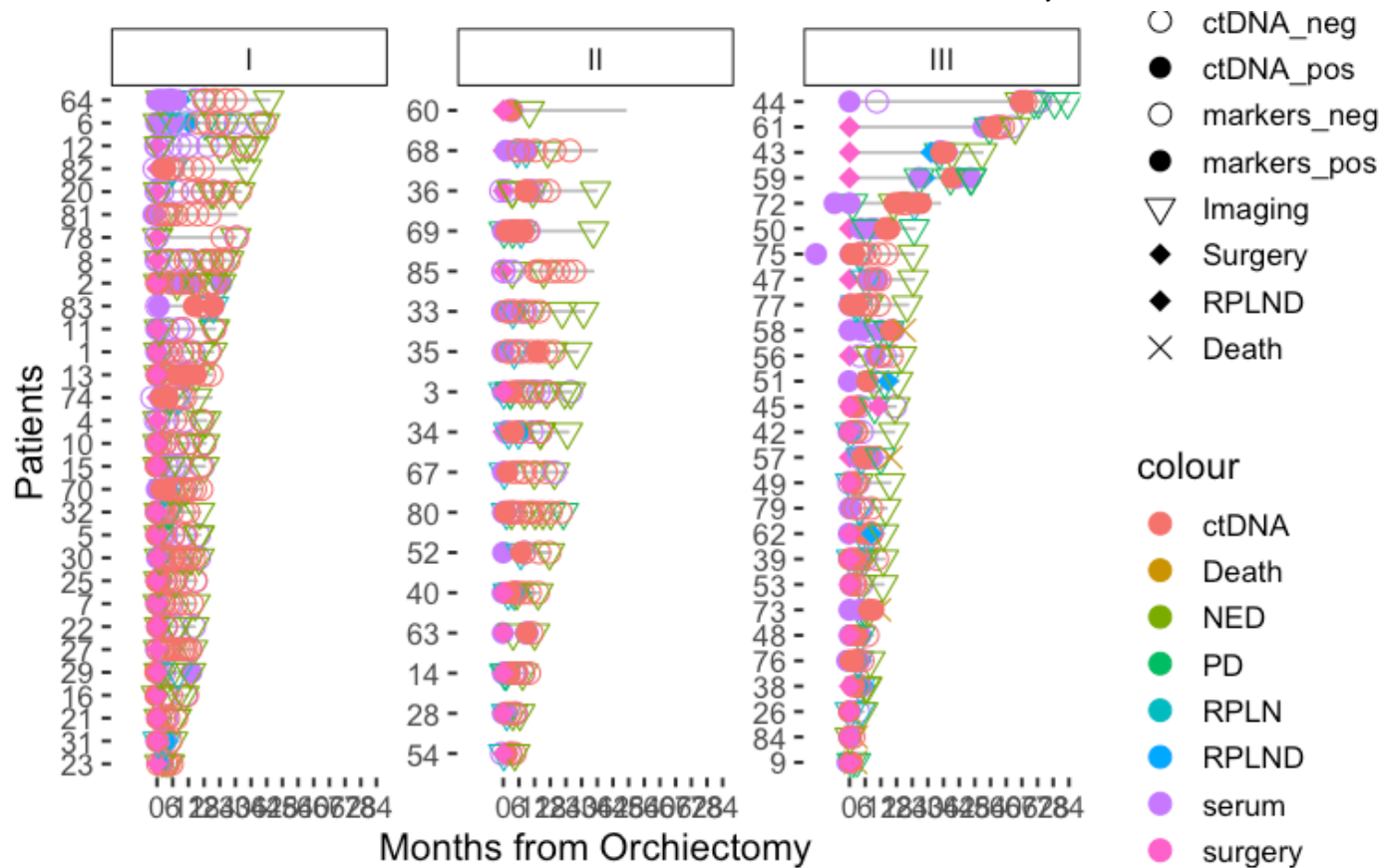


Hide

```
#Shape customization to Event_type
```

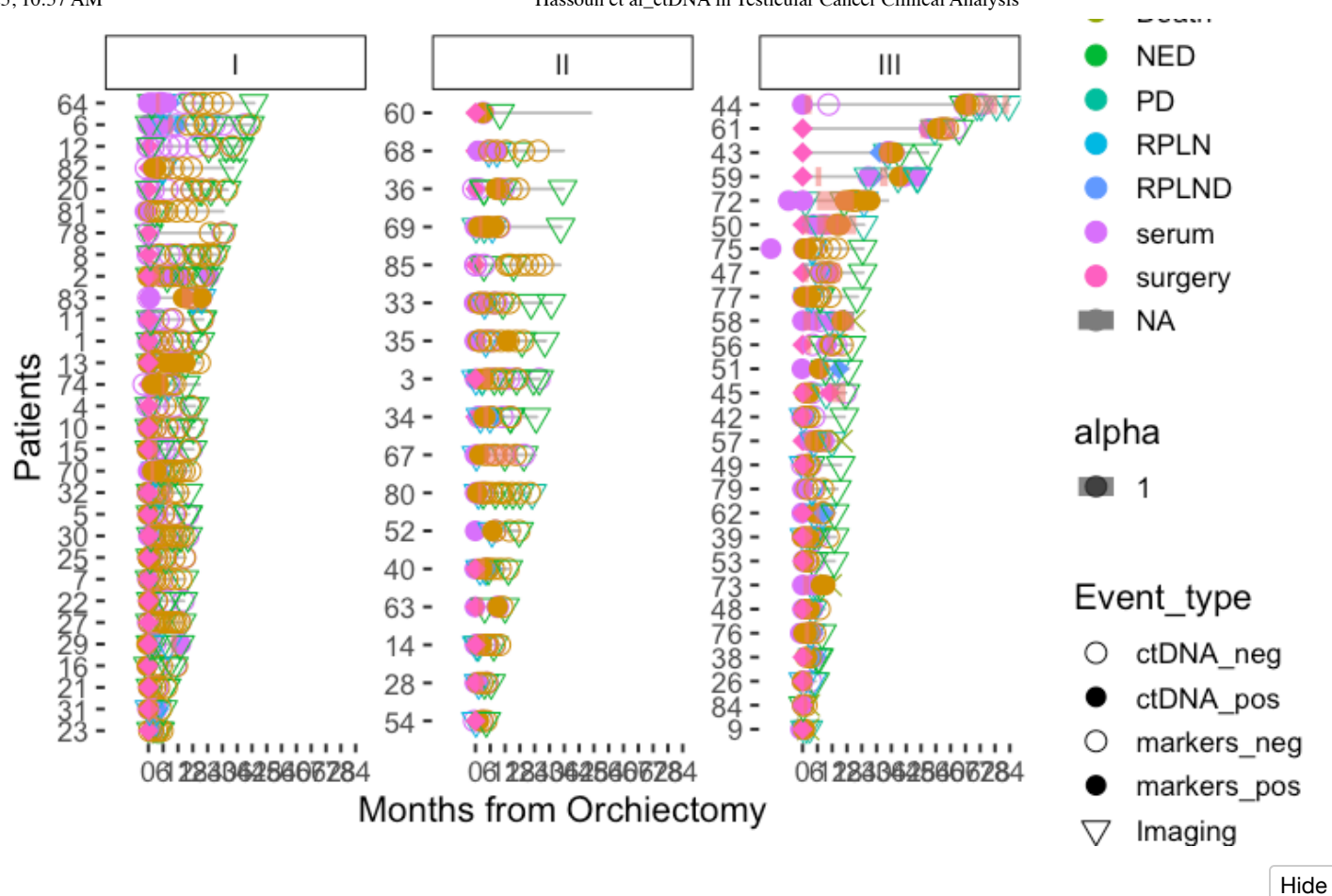
```
oplot_ev3.1 <- oplot_ev3 + ggplot2::scale_shape_manual(name="Event_type", values=c(1,16,
1,16,6,18,18,4), breaks=c('ctDNA_neg','ctDNA_pos','markers_neg','markers_pos','Imagin
g','Surgery','RPLND','Death'))
```

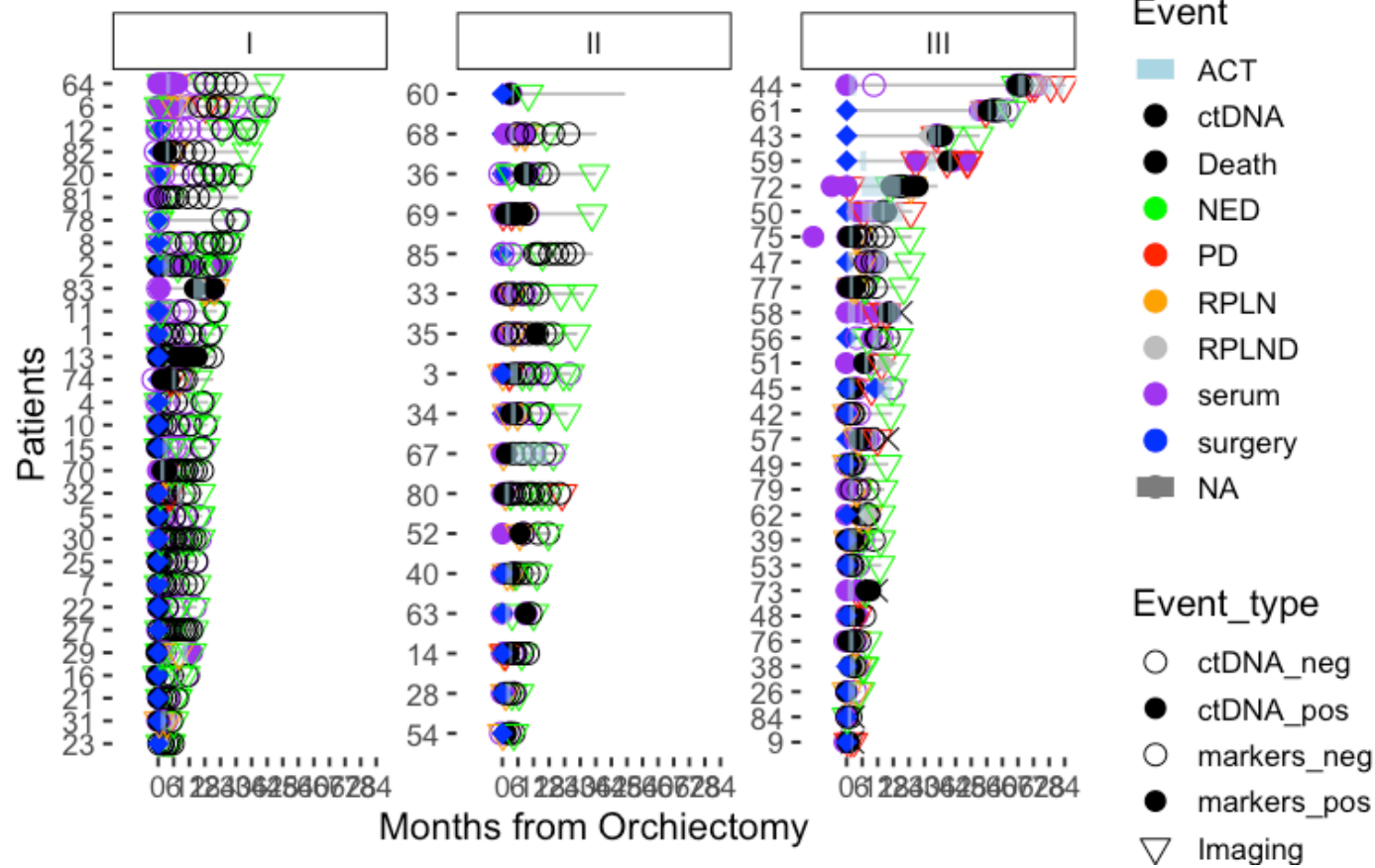
```
oplot_ev3.1
```


[Hide](#)

```
#plot treatment
oplot_ev4 <- oplot_ev3.1 + swimmer_lines(df_lines=clinstage_df,
                                         id='PatientName',
                                         start='Tx_start.months',
                                         end='Tx_end.months',
                                         name_col='Tx_type',
                                         size=3.5,
                                         name_alpha = 1.0)

oplot_ev4 <- oplot_ev4 + guides(linetype = guide_legend(override.aes = list(size = 5, color = "black")))
oplot_ev4
```





#Sankey Plots for all pts with post-orchietomy timepoint available

Hide

```
##To run this commands, please visit: https://sankeymatic.com/build/
#No pre-orch [20] MRD positive #E67272
#No pre-orch [12] MRD negative #87EA86
#No pre-orch [27] no MRD timepoint #808080
#Pre-orch Positive [1] MRD positive #E67272
#Pre-orch Positive [8] MRD negative #87EA86
#Pre-orch Positive [5] no MRD timepoint #808080
#Pre-orch Negative [1] MRD negative #87EA86

#MRD positive [7] Surveillance positive #E67272
#MRD positive [7] Surveillance negative #87EA86
#MRD positive [7] no Surveillance timepoint #808080
#MRD negative [1] Surveillance positive #E67272
#MRD negative [18] Surveillance negative #87EA86
#MRD negative [2] no Surveillance timepoint #808080
#no MRD timepoint [5] Surveillance positive #E67272
#no MRD timepoint [13] Surveillance negative #87EA86
#no MRD timepoint [14] no Surveillance timepoint #808080

#Surveillance positive [13] Event #E67272
#Surveillance positive [0] No Event #87EA86
#Surveillance negative [5] Event #E67272
#Surveillance negative [33] No Event #87EA86
#no Surveillance timepoint [17] Event #E67272
#no Surveillance timepoint [6] No Event #87EA86
```

#EFS by ctDNA at the MRD Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.MRD.Window!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.MRD.Window,
data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$PFS.months, event = circ_data\$PFS.Event) ~
ctDNA.MRD.Window, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.MRD.Window=NEGATIVE	21	3	NA	NA	NA
ctDNA.MRD.Window=POSITIVE	21	13	5.36	4.07	NA

Hide

```
event_summary <- circ_data %>%
  group_by(ctDNA.MRD.Window) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

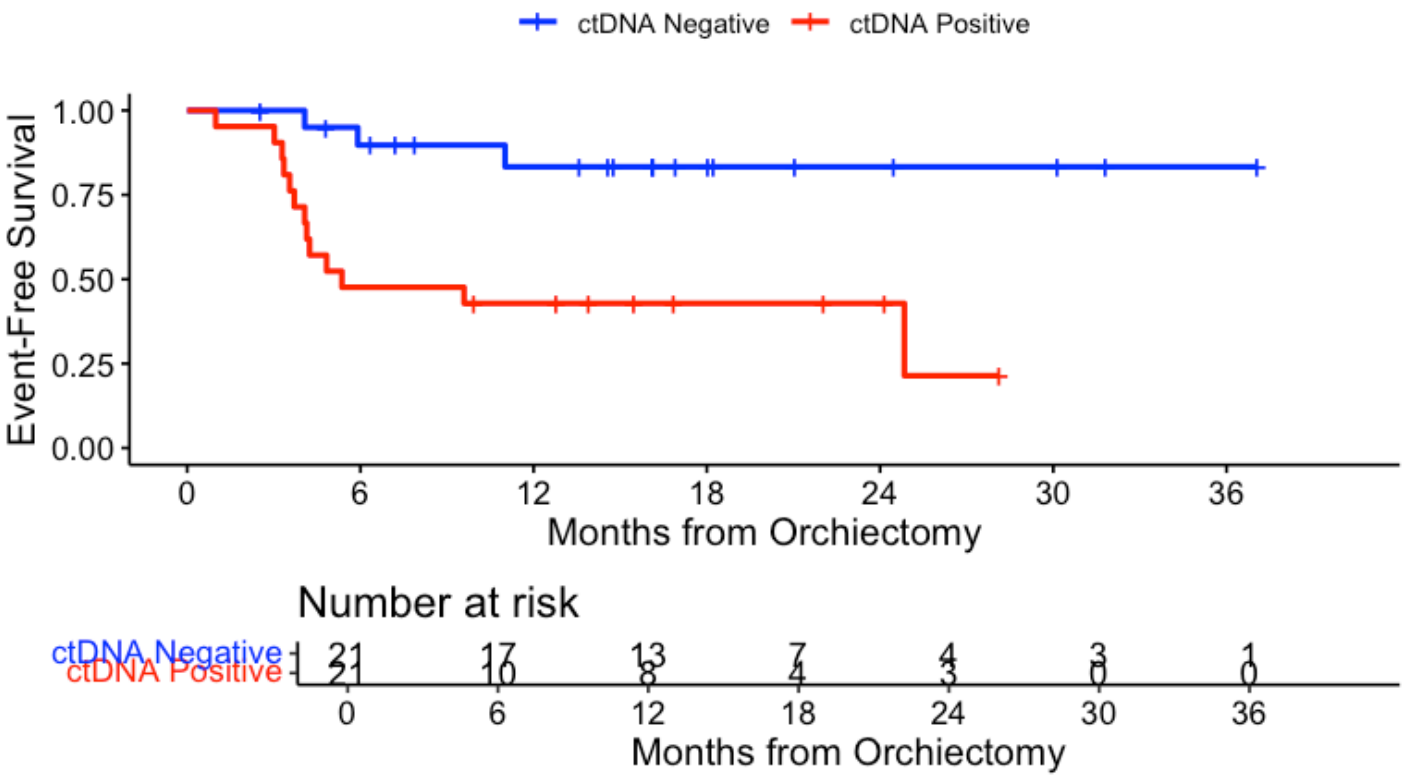
ctDNA.MRD.Window <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	21	3	0.1428571	14.28571
POSITIVE	21	13	0.6190476	61.90476

2 rows

Hide

```
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.MRD.Window, data = circ_data, conf.int=0.95, conf.
type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE,
break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA MRD window", ylab= "Event-Free Survival",
xlab="Months from Orchiectomy", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```


EFS - ctDNA MRD window



Hide

```
summary(KM_curve, times= c(12, 24))
```

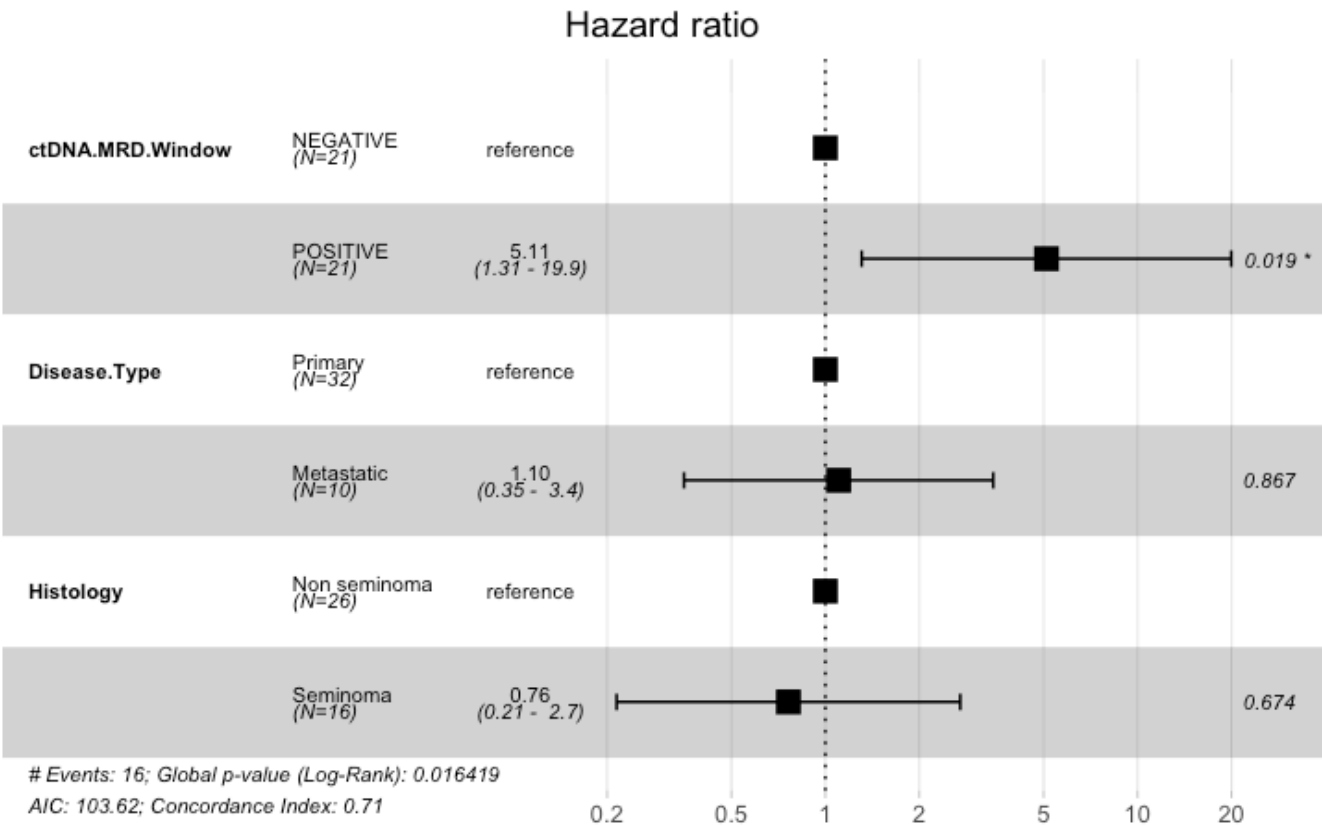
```
Call: survfit(formula = surv_object ~ ctDNA.MRD.Window, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

ctDNA.MRD.Window=NEGATIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	95% CI
12	13	3	0.833	0.0889	0.563	0.944	
24	4	0	0.833	0.0889	0.563	0.944	

ctDNA.MRD.Window=POSITIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	95% CI
12	8	12	0.429	0.108	0.219	0.623	
24	3	0	0.429	0.108	0.219	0.623	

Hide

```
circ_data$ctDNA.MRD.Window <- factor(circ_data$ctDNA.MRD.Window, levels=c("NEGATIVE","POSITIVE"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastatic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminoma"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ ctDNA.MRD.Window + Disease.Type + Histology, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.MRD.Window + Disease.Type +
      Histology, data = circ_data)
```

n= 42, number of events= 16

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.MRD.WindowPOSITIVE	1.63048	5.10634	0.69515	2.346	0.019 *
Disease.TypeMetastatic	0.09773	1.10267	0.58170	0.168	0.867
HistologySeminoma	-0.27236	0.76158	0.64651	-0.421	0.674

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.MRD.WindowPOSITIVE	5.1063	0.1958	1.3074	19.945
Disease.TypeMetastatic	1.1027	0.9069	0.3526	3.448
HistologySeminoma	0.7616	1.3131	0.2145	2.704

Concordance= 0.707 (se = 0.053)

Likelihood ratio test= 10.27 on 3 df, p=0.02

Wald test = 7.8 on 3 df, p=0.05

Score (logrank) test = 9.96 on 3 df, p=0.02

#EFS by STMs at the MRD Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.MRD.Window!="",]
circ_data <- circ_data[circ_data$Markers.MRD.Window!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~Markers.MRD.Window, data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$PFS.months, event = circ_data\$PFS.Event) ~ Markers.MRD.Window, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
Markers.MRD.Window=NEGATIVE	16	3	NA	NA	NA
Markers.MRD.Window=POSITIVE	22	11	24.8	4.14	NA

Hide

```
event_summary <- circ_data %>%
  group_by(Markers.MRD.Window) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

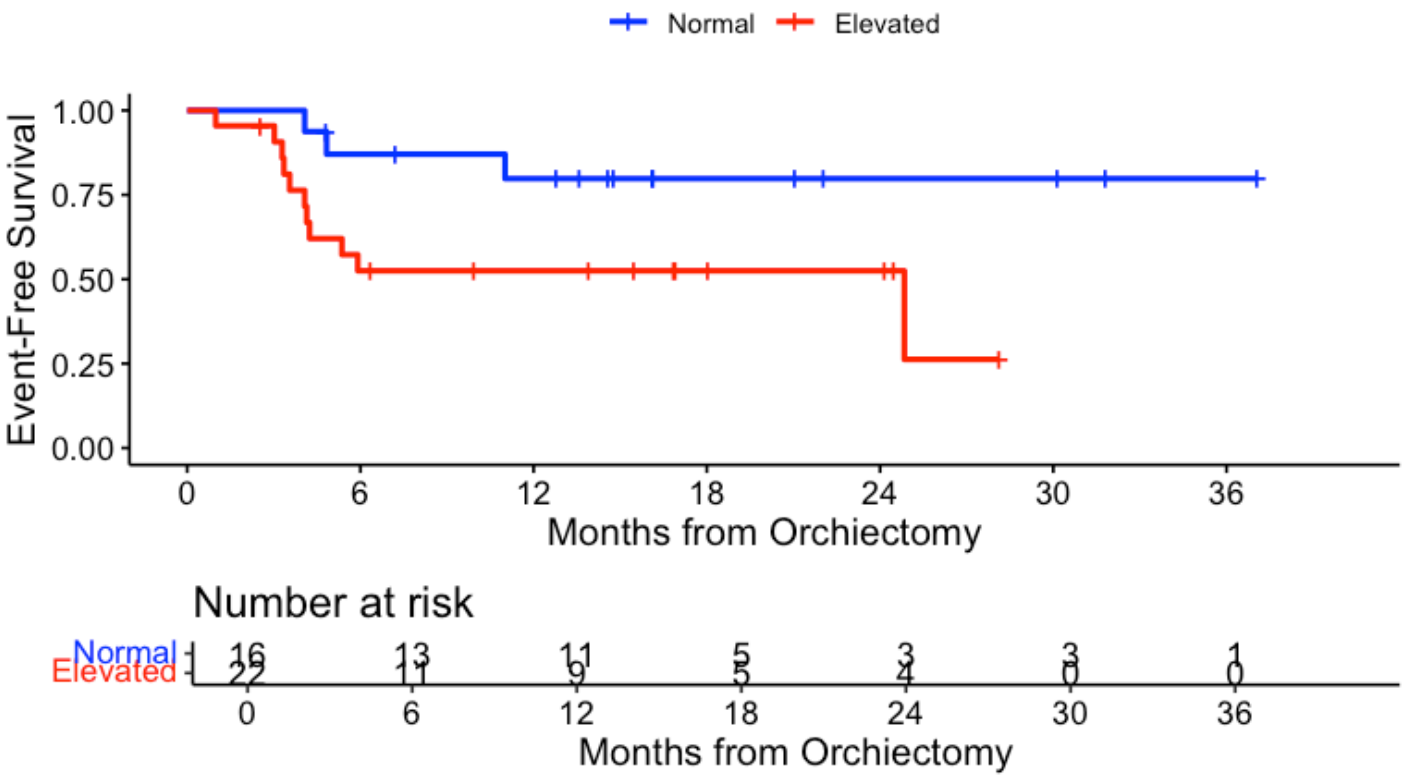
Markers.MRD.Window	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
NEGATIVE	16	3	0.1875	18.75
POSITIVE	22	11	0.5000	50.00

2 rows

Hide

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ Markers.MRD.Window, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - STM at MRD window", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("Normal", "Elevated"), legend.title="")
```

EFS - STM at MRD window



Hide

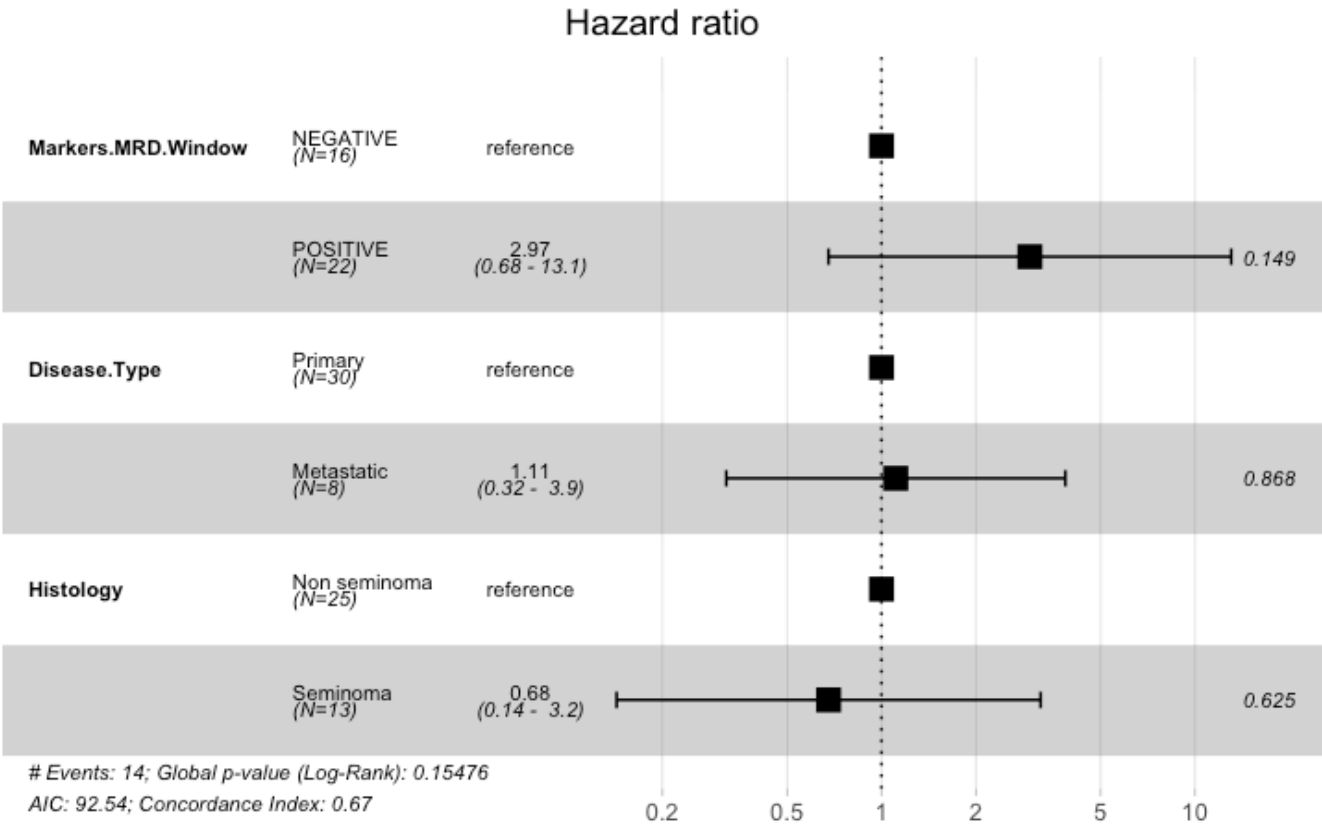
```
summary(KM_curve, times= c(12))
```

Call: `survfit(formula = surv_object ~ Markers.MRD.Window, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

Markers.MRD.Window=NEGATIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
30	12.000	11.000	3.000	0.798	0.105	0.494	0.9
Markers.MRD.Window=POSITIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
10	12.000	9.000	10.000	0.525	0.109	0.298	0.7

Hide

```
circ_data$Markers.MRD.Window <- factor(circ_data$Markers.MRD.Window, levels=c("NEGATIV
E","POSITIVE"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastat
ic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminom
a"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ Markers.MRD.Window + Disease.Type + Histology, data=circ_
data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ Markers.MRD.Window + Disease.Type +
      Histology, data = circ_data)
```

n= 38, number of events= 14

	coef	exp(coef)	se(coef)	z	Pr(> z)
Markers.MRD.WindowPOSITIVE	1.0898	2.9738	0.7547	1.444	0.149
Disease.TypeMetastatic	0.1053	1.1110	0.6353	0.166	0.868
HistologySeminoma	-0.3885	0.6781	0.7945	-0.489	0.625

	exp(coef)	exp(-coef)	lower .95	upper .95
Markers.MRD.WindowPOSITIVE	2.9738	0.3363	0.6775	13.054
Disease.TypeMetastatic	1.1110	0.9001	0.3198	3.859
HistologySeminoma	0.6781	1.4747	0.1429	3.218

Concordance= 0.669 (se = 0.061)

Likelihood ratio test= 5.24 on 3 df, p=0.2

Wald test = 4.29 on 3 df, p=0.2

Score (logrank) test = 4.91 on 3 df, p=0.2

#EFS by ctDNA/STMs combined at the MRD Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.MRD.Window!="",]
circ_data <- circ_data[circ_data$Markers.MRD.Window!="",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA #first we create the variable for the ctDNA & NAC combina
tion, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.MRD.Window == "NEGATIVE" & Markers.MRD.Window == "NEGATIVE" ~ "ctDNA/STMs
(-)",
    (ctDNA.MRD.Window == "POSITIVE" & Markers.MRD.Window == "NEGATIVE") |
    (ctDNA.MRD.Window == "NEGATIVE" & Markers.MRD.Window == "POSITIVE") ~ "Discordan
t",
    ctDNA.MRD.Window == "POSITIVE" & Markers.MRD.Window == "POSITIVE" ~ "ctDNA/STMs (+)"
  ))

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.Dynamics, d
ata = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
  ctDNA.Dynamics, data = circ_data)
```

	n	events	median	0.95LCL	0.95UCL
ctDNA.Dynamics=ctDNA/STMs (-)	13	2	NA	NA	NA
ctDNA.Dynamics=ctDNA/STMs (+)	16	10	4.8	3.55	NA
ctDNA.Dynamics=Discordant	9	2	NA	NA	NA

Hide

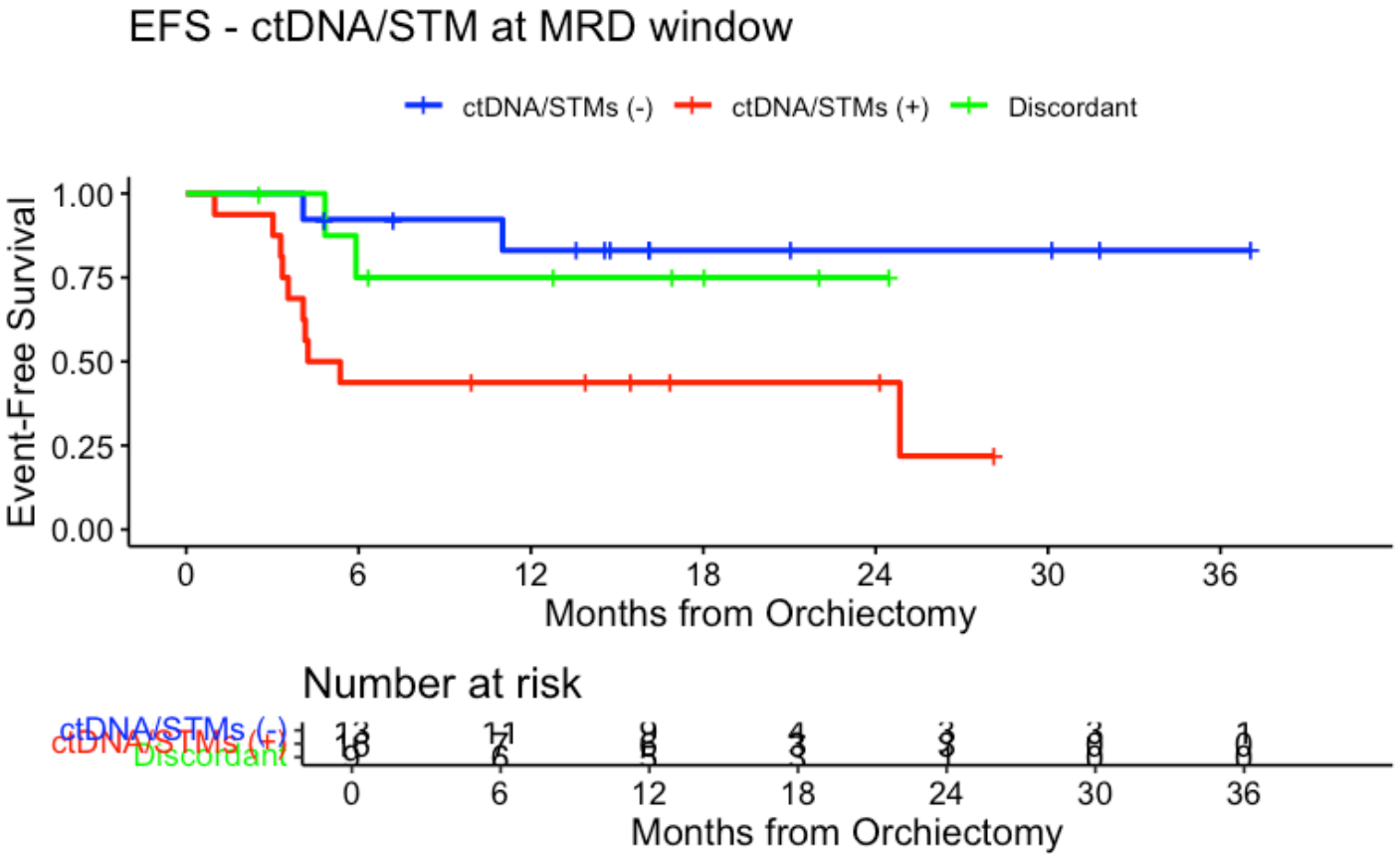
```
event_summary <- circ_data %>%
  group_by(ctDNA.Dynamics) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.Dynamics <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
Discordant	9	2	0.2222222	22.22222
ctDNA/STMs (+)	16	10	0.6250000	62.50000
ctDNA/STMs (-)	13	2	0.1538462	15.38462

3 rows

Hide

```
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.Dynamics, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red","green"), title="EFS - ctDNA/STM at MRD window", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("ctDNA/STMs (-)","ctDNA/STMs (+)","Discordant"), legend.title="")
```

Hide

```
summary(KM_curve, times= c(12))
```

```
Call: survfit(formula = surv_object ~ ctDNA.Dynamics, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

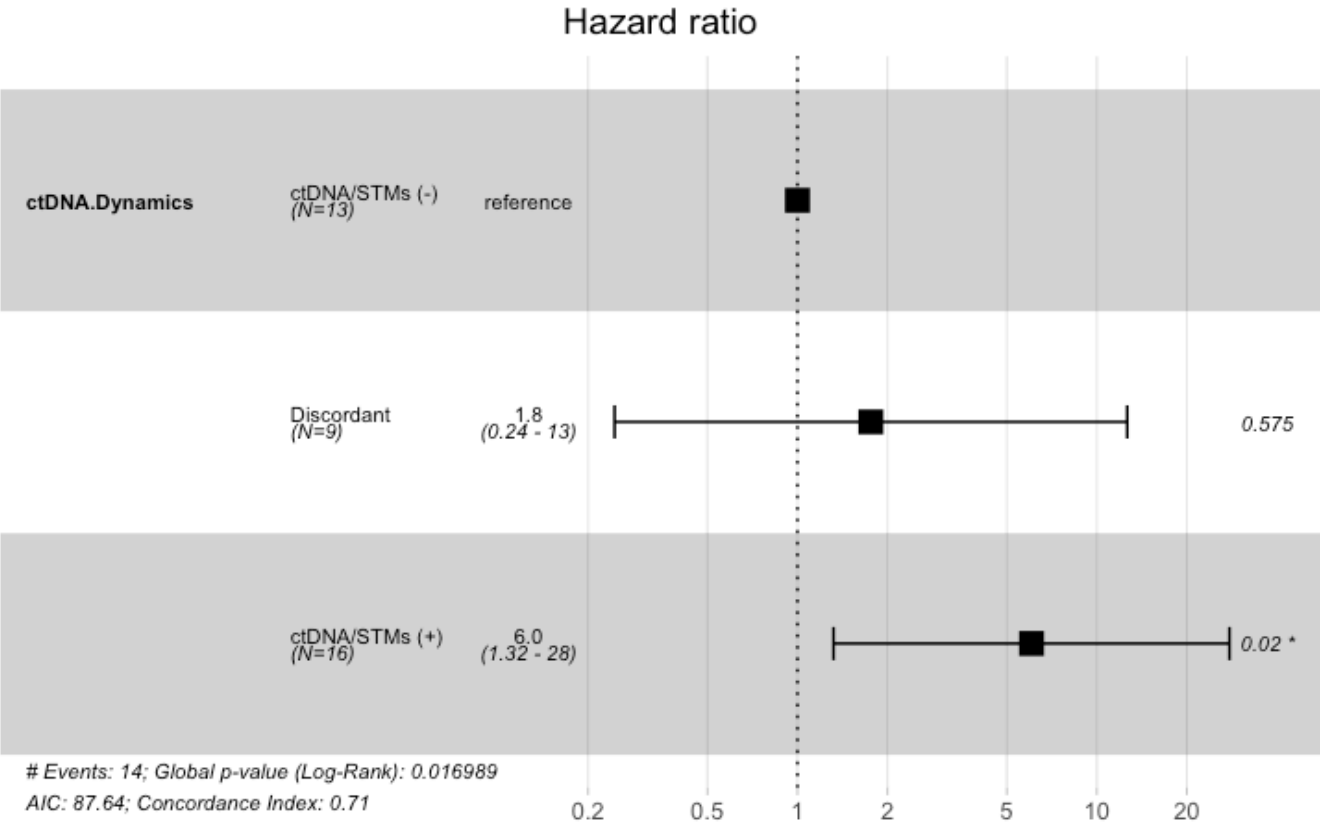
		ctDNA.Dynamics=ctDNA/STMs (-)					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
55	12.000	9.000	2.000	0.831	0.110	0.472	0.9

		ctDNA.Dynamics=ctDNA/STMs (+)					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
56	12.000	6.000	9.000	0.438	0.124	0.198	0.6

		ctDNA.Dynamics=Discordant					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
31	12.000	5.000	2.000	0.750	0.153	0.315	0.9

[Hide](#)

```
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("ctDNA/STMs
(-)","Discordant","ctDNA/STMs (+)"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastatic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminoma"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:
coxph(formula = surv_object ~ ctDNA.Dynamics, data = circ_data)

n= 38, number of events= 14

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.DynamicsDiscordant	0.5646	1.7588	1.0057	0.561	0.5745
ctDNA.DynamicsctDNA/STMs (+)	1.7998	6.0483	0.7767	2.317	0.0205 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.DynamicsDiscordant	1.759	0.5686	0.245	12.63
ctDNA.DynamicsctDNA/STMs (+)	6.048	0.1653	1.320	27.72

Concordance= 0.713 (se = 0.06)
Likelihood ratio test= 8.15 on 2 df, p=0.02
Wald test = 6.84 on 2 df, p=0.03
Score (logrank) test = 8.47 on 2 df, p=0.01

#Median MTM/mL at the MRD Window and by PFS status

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available == TRUE, ]
circ_data <- circ_data[circ_data$ctDNA.MRD.Window != "", ]

# 1. Median and range of ctDNA.MRD.MTM for POSITIVE MRD Window
positive_data <- circ_data %>% filter(ctDNA.MRD.Window == "POSITIVE")
median_positive <- median(positive_data$ctDNA.MRD.MTM, na.rm = TRUE)
range_positive <- range(positive_data$ctDNA.MRD.MTM, na.rm = TRUE)
cat("Median ctDNA.MRD.MTM in MRD-Positive group:", median_positive, "\n")
```

Median ctDNA.MRD.MTM in MRD-Positive group: 50.53

Hide

```
cat("Range of ctDNA.MRD.MTM in MRD-Positive group:", range_positive[1], "-", range_positive[2], "\n\n")
```

Range of ctDNA.MRD.MTM in MRD-Positive group: 3.15 – 756.65

#EFS by ctDNA at the Surveillance Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.surveillance!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.surveillance, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
ctDNA.surveillance, data = circ_data)
```

	n	events	median	0.95LCL	0.95UCL
ctDNA.surveillance=NEGATIVE	38	5	NA	NA	NA
ctDNA.surveillance=POSITIVE	13	13	4.83	3.35	NA

Hide

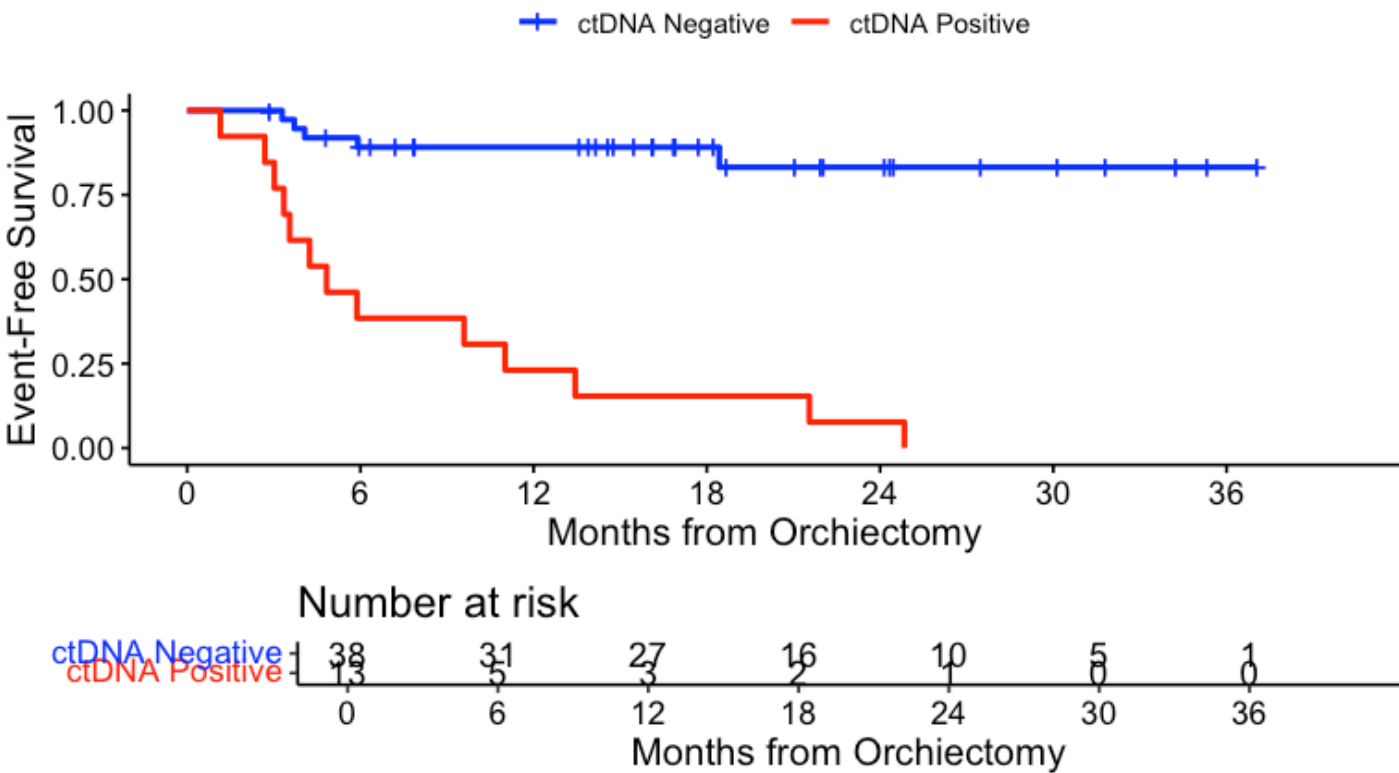
```
event_summary <- circ_data %>%
  group_by(ctDNA.surveillance) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.surveillance	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
NEGATIVE	38	5	0.1315789	13.15789
POSITIVE	13	13	1.0000000	100.00000
2 rows				

[Hide](#)

```
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.surveillance, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA Surveillance window", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```

EFS - ctDNA Surveillance window



Hide

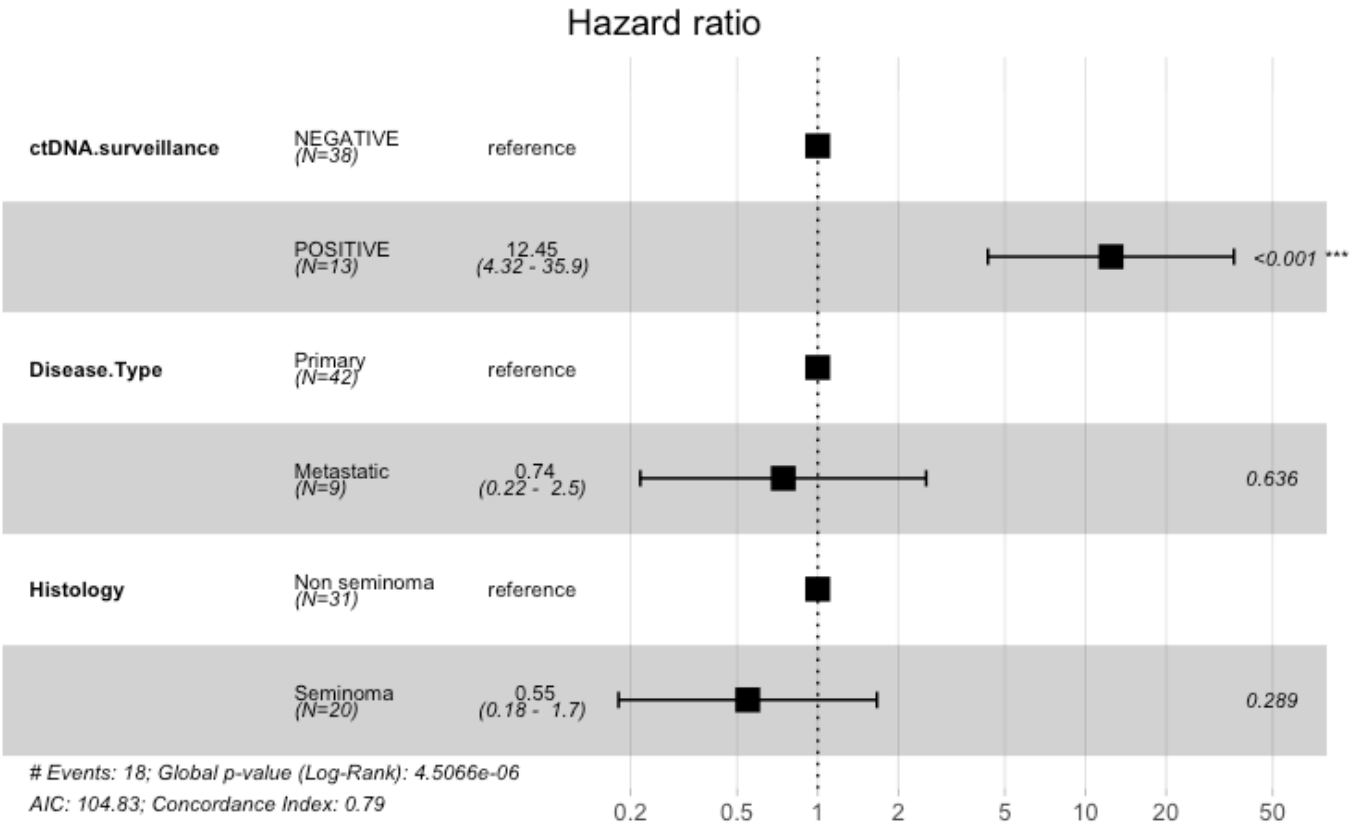
```
summary(KM_curve, times= c(12))
```

Call: `survfit(formula = surv_object ~ ctDNA.surveillance, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

ctDNA.surveillance=NEGATIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI	12.0000	27.0000	4.0000	0.8911	0.0514	0.7352	0.95
77							
ctDNA.surveillance=POSITIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI	12.0000	3.0000	10.0000	0.2308	0.1169	0.0558	0.47
46							

Hide

```
circ_data$ctDNA.surveillance <- factor(circ_data$ctDNA.surveillance, levels=c("NEGATIVE", "POSITIVE"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastatic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminoma"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ ctDNA.surveillance + Disease.Type + Histology, data=circ_data)
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.surveillance + Disease.Type +
      Histology, data = circ_data)
```

n= 51, number of events= 18

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.surveillancePOSITIVE	2.5216	12.4485	0.5397	4.672	2.98e-06 ***
Disease.TypeMetastatic	-0.2967	0.7433	0.6270	-0.473	0.636
HistologySeminoma	-0.6017	0.5479	0.5670	-1.061	0.289

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.surveillancePOSITIVE	12.4485	0.08033	4.3222	35.854
Disease.TypeMetastatic	0.7433	1.34535	0.2175	2.540
HistologySeminoma	0.5479	1.82529	0.1803	1.665

Concordance= 0.793 (se = 0.069)

Likelihood ratio test= 27.55 on 3 df, p=5e-06

Wald test = 23.95 on 3 df, p=3e-05

Score (logrank) test = 38.6 on 3 df, p=2e-08

#EFS by Serum Markers at the Surveillance Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$Markers.Surveillance!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~Markers.Surveillance, data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$PFS.months, event = circ_data\$PFS.Event) ~ Markers.Surveillance, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
Markers.Surveillance=NEGATIVE	48	15	65.3	NA	NA
Markers.Surveillance=POSITIVE	14	11	18.4	7.82	NA

Hide


```
event_summary <- circ_data %>%
  group_by(Markers.Surveillance) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

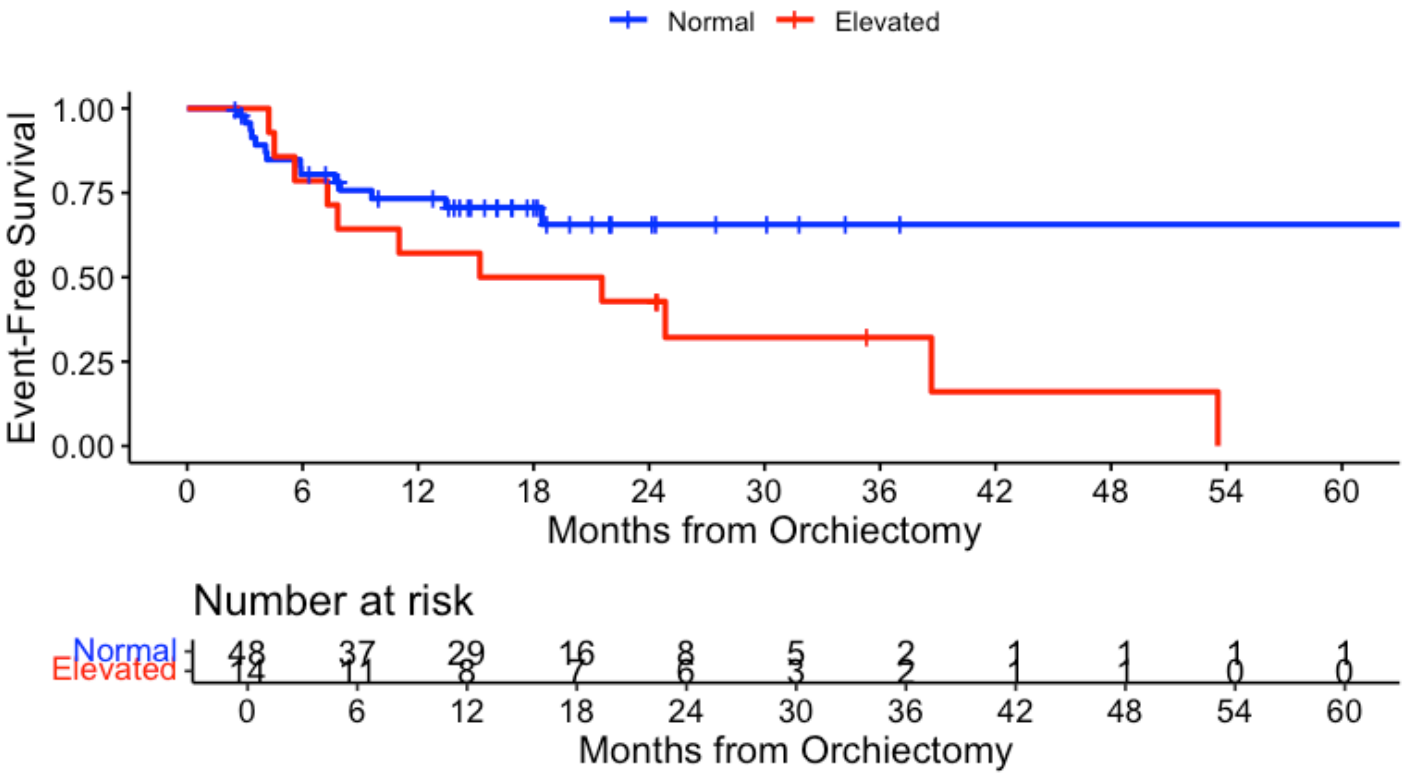
Markers.Surveillance	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
NEGATIVE	48	15	0.3125000	31.25000
POSITIVE	14	11	0.7857143	78.57143

2 rows

Hide

```
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ Markers.Surveillance, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - Serum Markers Surveillance window", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("Normal", "Elevated"), legend.title="")
```

EFS - Serum Markers Surveillance window



Hide

```
summary(KM_curve, times= c(12))
```

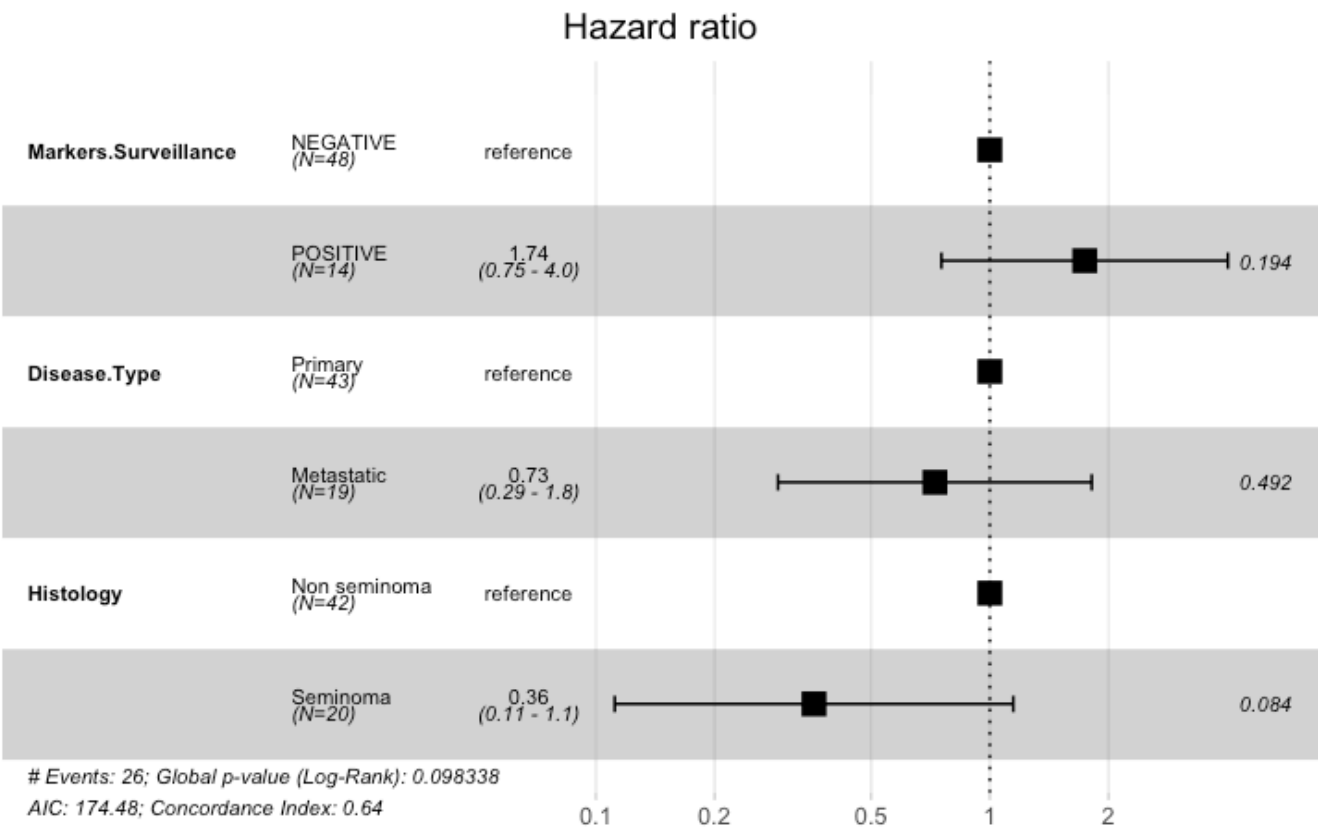
```
Call: survfit(formula = surv_object ~ Markers.Surveillance, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

Markers.Surveillance=NEGATIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
89	12.0000	29.0000	12.0000	0.7329	0.0663	0.5770	0.83

Markers.Surveillance=POSITIVE							
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
80	12.000	8.000	6.000	0.571	0.132	0.284	0.7

Hide

```
circ_data$Markers.Surveillance <- factor(circ_data$Markers.Surveillance, levels=c("NEGATIVE", "POSITIVE"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastatic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminoma"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ Markers.Surveillance + Disease.Type + Histology, data=circ_data)
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ Markers.Surveillance + Disease.Type +
      Histology, data = circ_data)
```

n= 62, number of events= 26

	coef	exp(coef)	se(coef)	z	Pr(> z)
Markers.SurveillancePOSITIVE	0.5545	1.7410	0.4271	1.298	0.1942
Disease.TypeMetastatic	-0.3215	0.7251	0.4678	-0.687	0.4919
HistologySeminoma	-1.0279	0.3578	0.5944	-1.729	0.0838 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Markers.SurveillancePOSITIVE	1.7410	0.5744	0.7538	4.021
Disease.TypeMetastatic	0.7251	1.3792	0.2899	1.814
HistologySeminoma	0.3578	2.7952	0.1116	1.147

Concordance= 0.638 (se = 0.05)

Likelihood ratio test= 6.29 on 3 df, p=0.1

Wald test = 5.69 on 3 df, p=0.1

Score (logrank) test = 6.19 on 3 df, p=0.1

#EFS by ctDNA/STMs combined at the Surveillance Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ctDNA.surveillance!="",]
circ_data <- circ_data[circ_data$Markers.Surveillance!="",]
circ_datadf <- as.data.frame(circ_data)

circ_data$ctDNA.Dynamics <- NA #first we create the variable for the ctDNA & NAC combina
tion, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.surveillance == "NEGATIVE" & Markers.Surveillance == "NEGATIVE" ~ "ctDNA/STMs
(-)",
    (ctDNA.surveillance == "POSITIVE" & Markers.Surveillance == "NEGATIVE") |
    (ctDNA.surveillance == "NEGATIVE" & Markers.Surveillance == "POSITIVE") ~ "Discord
ant",
    ctDNA.surveillance == "POSITIVE" & Markers.Surveillance == "POSITIVE" ~ "ctDNA/STMs
(+)"
  ))

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.Dynamics, d
ata = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
  ctDNA.Dynamics, data = circ_data)
```

	n	events	median	0.95LCL	0.95UCL
ctDNA.Dynamics=ctDNA/STMs (-)	32	4	NA	NA	NA
ctDNA.Dynamics=ctDNA/STMs (+)	4	4	16.28	4.24	NA
ctDNA.Dynamics=Discordant	10	7	7.73	3.35	NA

Hide

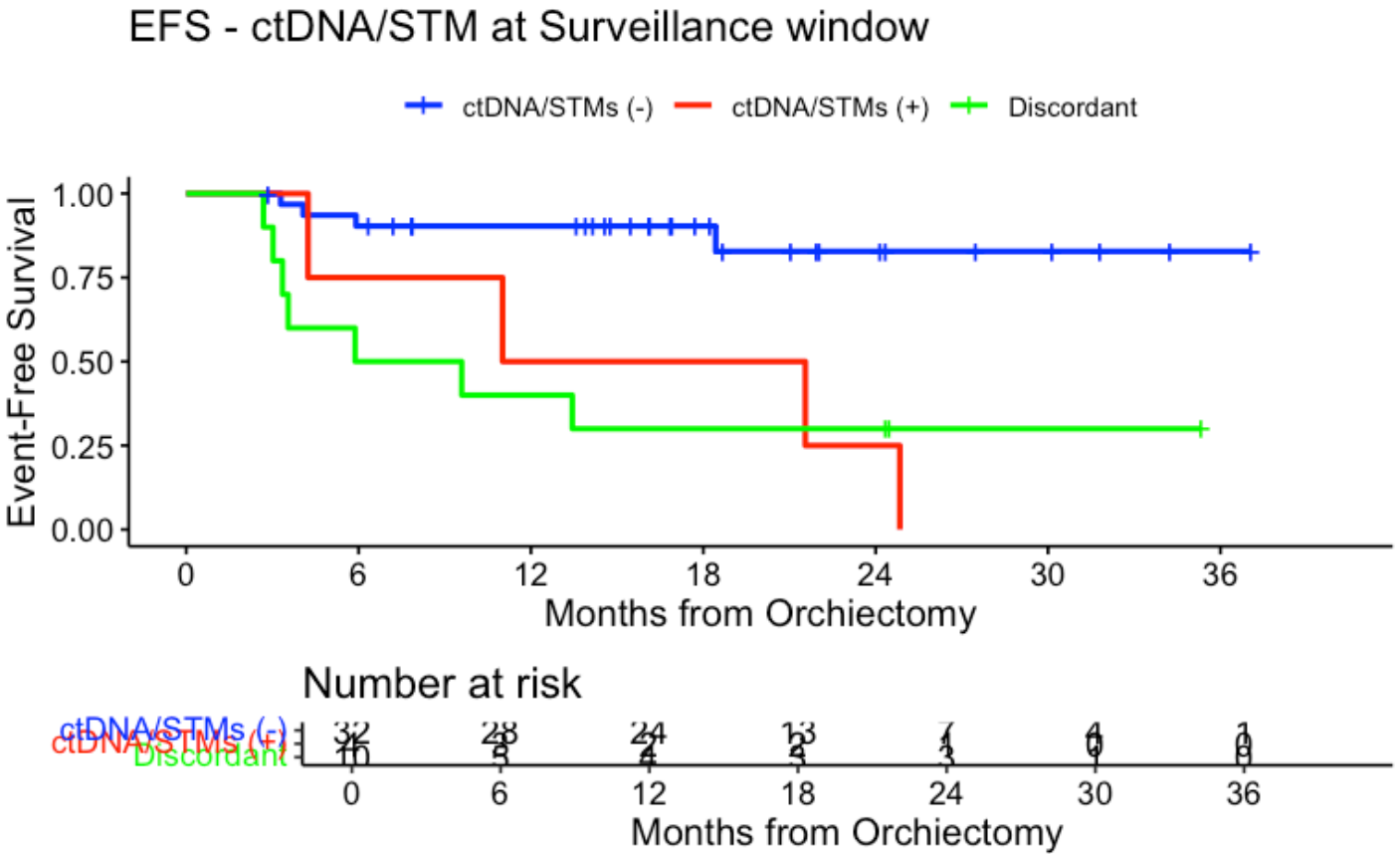
```
event_summary <- circ_data %>%
  group_by(ctDNA.Dynamics) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.Dynamics	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
Discordant	10	7	0.700	70.0
ctDNA/STMs (+)	4	4	1.000	100.0
ctDNA/STMs (-)	32	4	0.125	12.5

3 rows

Hide

```
surv_object <- Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.Dynamics, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red","green"), title="EFS - ctDNA/STM at Surveillance window", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("ctDNA/STMs (-)","ctDNA/STMs (+)","Discordant"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(12))
```

```
Call: survfit(formula = surv_object ~ ctDNA.Dynamics, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

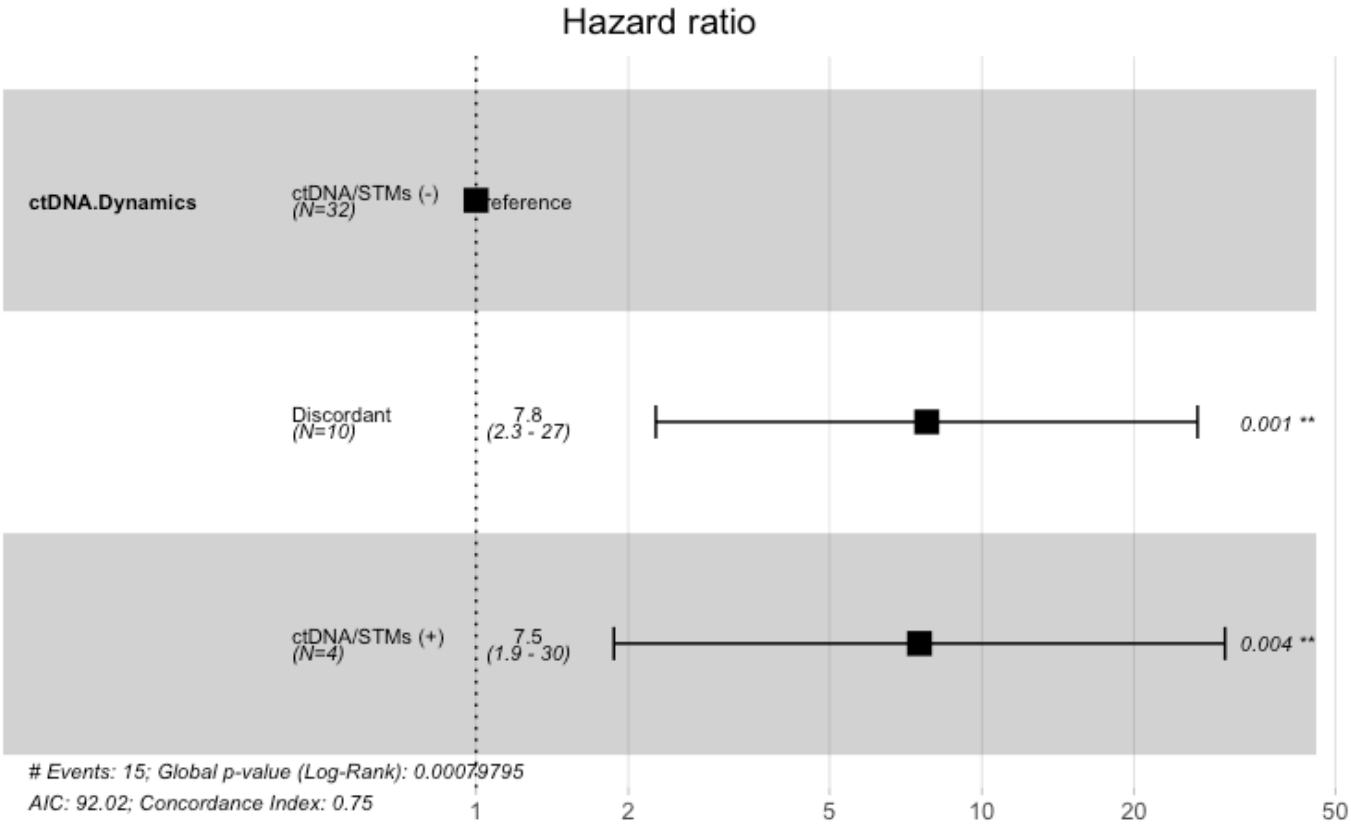
		ctDNA.Dynamics=ctDNA/STMs (-)					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
77	12.0000	24.0000	3.0000	0.9032	0.0531	0.7293	0.96

		ctDNA.Dynamics=ctDNA/STMs (+)					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
49	12.0000	2.0000	2.0000	0.5000	0.2500	0.0578	0.84

		ctDNA.Dynamics=Discordant					
	time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95%
CI							
70	12.000	4.000	6.000	0.400	0.155	0.123	0.6

[Hide](#)

```
circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("ctDNA/STMs
(-)","Discordant","ctDNA/STMs (+)"))
circ_data$Disease.Type <- factor(circ_data$Disease.Type, levels = c("Primary", "Metastatic"))
circ_data$Histology <- factor(circ_data$Histology, levels = c("Non seminoma", "Seminoma"), labels = c("Non seminoma", "Seminoma"))
cox_fit <- coxph(surv_object ~ ctDNA.Dynamics, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:
coxph(formula = surv_object ~ ctDNA.Dynamics, data = circ_data)

n= 46, number of events= 15

	coef	exp(coef)	se(coef)	z	Pr(> z)
ctDNA.DynamicsDiscordant	2.0516	7.7803	0.6292	3.261	0.00111 **
ctDNA.DynamicsctDNA/STMs (+)	2.0185	7.5272	0.7099	2.844	0.00446 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ctDNA.DynamicsDiscordant	7.780	0.1285	2.267	26.70
ctDNA.DynamicsctDNA/STMs (+)	7.527	0.1329	1.872	30.26

Concordance= 0.749 (se = 0.065)
Likelihood ratio test= 14.27 on 2 df, p=8e-04
Wald test = 12.11 on 2 df, p=0.002
Score (logrank) test = 16.8 on 2 df, p=2e-04

#EFS by ctDNA at post-RPLND - Stage II

Hide


```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$RPLND==TRUE,]
circ_data <- circ_data[circ_data$Stage=="II",]
circ_data <- circ_data[circ_data$ctDNA.post.RPLND!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.post.RPLND,
data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$PFS.months, event = circ_data\$PFS.Event) ~
ctDNA.post.RPLND, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.post.RPLND=NEGATIVE	4	2	7.69	5.91	NA
ctDNA.post.RPLND=POSITIVE	2	2	1.92	1.15	NA

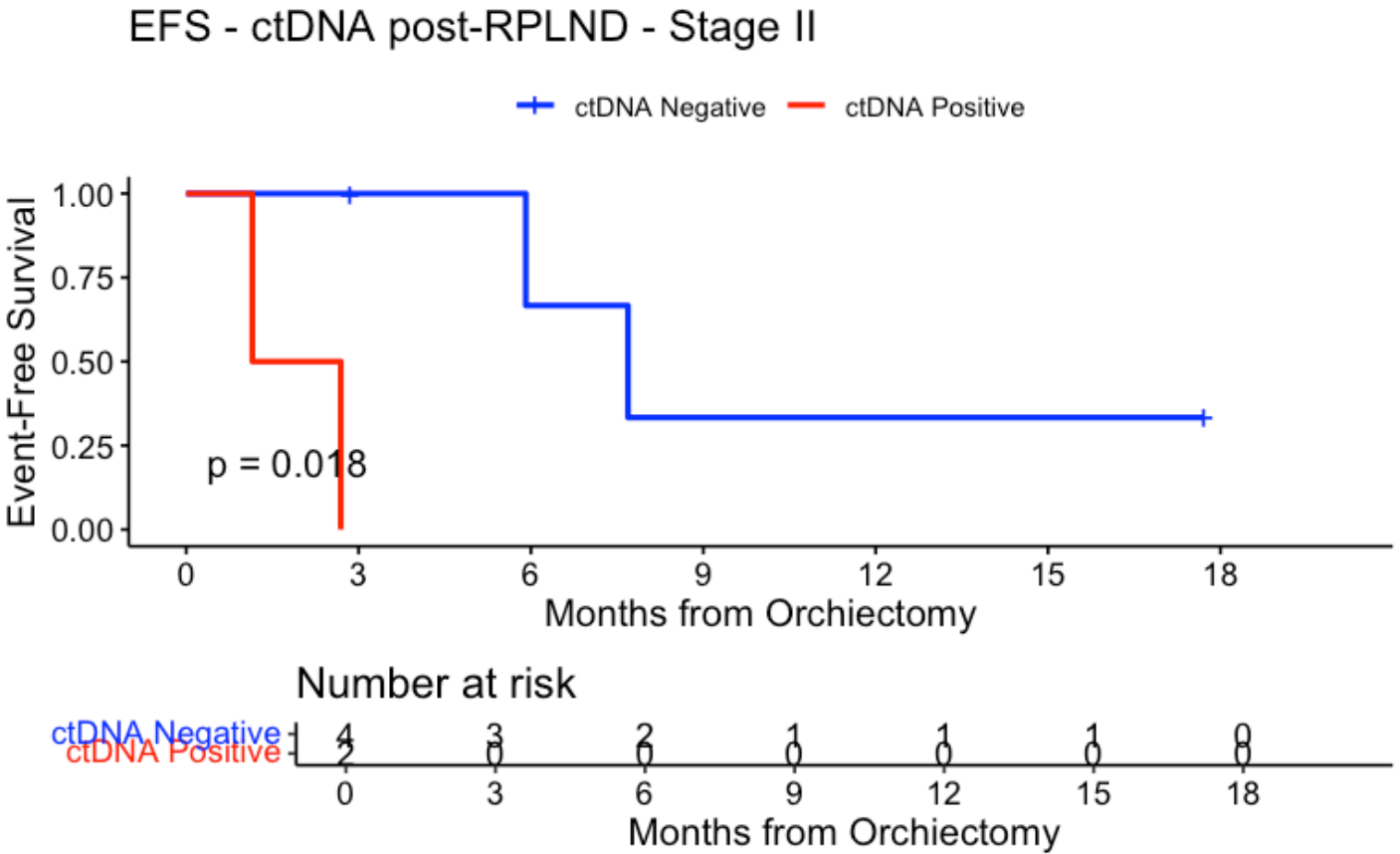
[Hide](#)

```
event_summary <- circ_data %>%
  group_by(ctDNA.post.RPLND) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.post.RPLND <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	4	2	0.5	50
POSITIVE	2	2	1.0	100
2 rows				

[Hide](#)

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.post.RPLND, data = circ_data,conf.int=0.95,conf.
type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = TRUE, conf.int = FALSE, risk.table = TRUE,
break.time.by=3, palette=c("blue","red"), title="EFS - ctDNA post-RPLND - Stage II", yla
b= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("ctDNA Negativ
e", "ctDNA Positive"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(0, 12))
```

```
Call: survfit(formula = surv_object ~ ctDNA.post.RPLND, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")

      ctDNA.post.RPLND=NEGATIVE
time  n.risk  n.event  survival  std.err  lower 95% CI  upper 95% CI
  0         4         0    1.000    0.000    1.00000    1.000
 12         1         2    0.333    0.272    0.00896    0.774

      ctDNA.post.RPLND=POSITIVE
CI      time      n.risk      n.event      survival      std.err  lower 95% CI  upper 95%
1         0         2         0         1         0         1
```

Hide

```
circ_data$ctDNA.post.RPLND <- factor(circ_data$ctDNA.post.RPLND, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxphf(surv_object ~ ctDNA.post.RPLND, data=circ_data)
summary(cox_fit)
```

```
coxphf(formula = surv_object ~ ctDNA.post.RPLND, data = circ_data)

Model fitted by Penalized ML
Confidence intervals and p-values by Profile Likelihood

      coef se(coef) exp(coef) lower 0.95 upper 0.95  Chisq
p
ctDNA.post.RPLNDPOSITIVE 2.715398 1.929072  15.11062   1.165602   2114.922 4.34548 0.03710731

Likelihood ratio test=4.34548 on 1 df, p=0.03710731, n=6
Wald test = 1.981391 on 1 df, p = 0.159244

Covariance-Matrix:
      ctDNA.post.RPLNDPOSITIVE
ctDNA.post.RPLNDPOSITIVE      3.721317
```

#EFS by ctDNA at post-ACT - Stage II/III

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ACT==TRUE,]
circ_data <- circ_data[circ_data$Stage!="I",]
circ_data <- circ_data[circ_data$ctDNA.surveillance!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~ctDNA.surveillance, data = circ_data)
```

Call: survfit(formula = Surv(time = circ_data\$PFS.months, event = circ_data\$PFS.Event) ~ ctDNA.surveillance, data = circ_data)

	n	events	median	0.95LCL	0.95UCL
ctDNA.surveillance=NEGATIVE	13	3	NA	NA	NA
ctDNA.surveillance=POSITIVE	9	9	4.24	3.35	NA

[Hide](#)

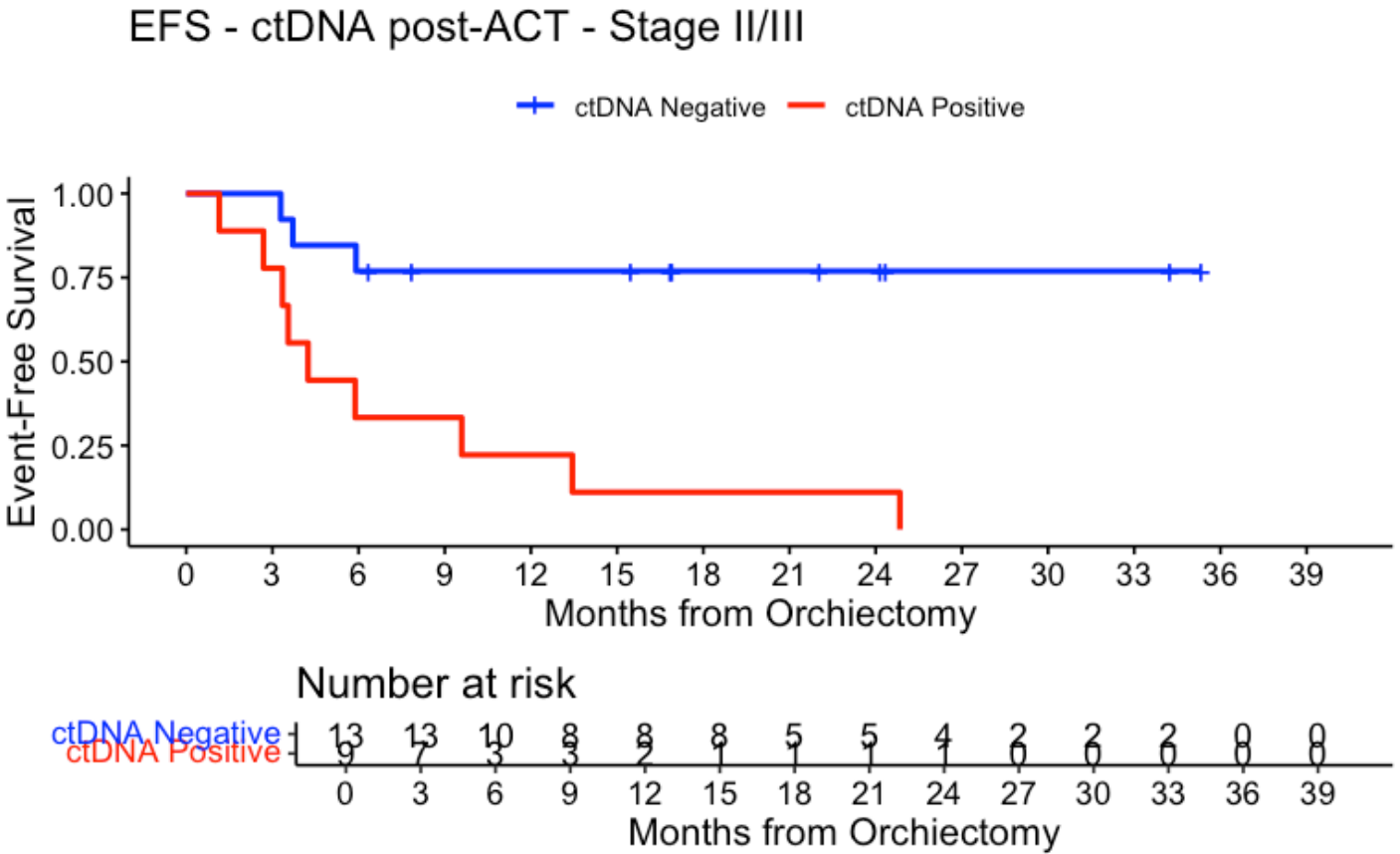
```
event_summary <- circ_data %>%
  group_by(ctDNA.surveillance) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

ctDNA.surveillance <chr>	Total <int>	Events <int>	Fraction <dbl>	Percentage <dbl>
NEGATIVE	13	3	0.2307692	23.07692
POSITIVE	9	9	1.0000000	100.00000

2 rows

[Hide](#)

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.surveillance, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=3, palette=c("blue","red"), title="EFS - ctDNA post-ACT - Stage II/II I", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(12, 24))
```

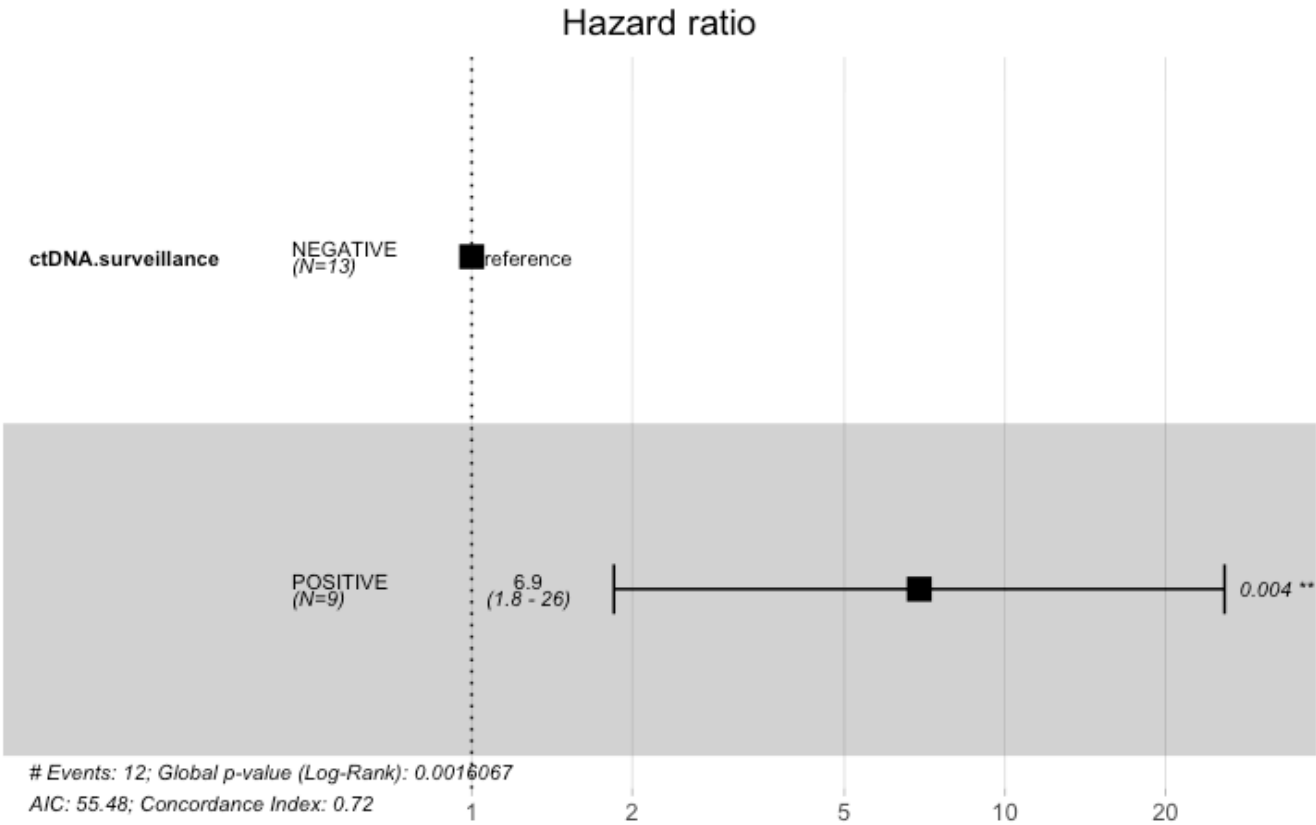
```
Call: survfit(formula = surv_object ~ ctDNA.surveillance, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

ctDNA.surveillance=NEGATIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
12	8	3	0.769	0.117	0.442	0.919	
24	4	0	0.769	0.117	0.442	0.919	

ctDNA.surveillance=POSITIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
12	2	7	0.222	0.139	0.03371	0.513	
24	1	1	0.111	0.105	0.00613	0.388	

Hide

```
circ_data$ctDNA.surveillance <- factor(circ_data$ctDNA.surveillance, levels=c("NEGATIVE", "POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.surveillance, data=circ_data)
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

```
Call:
coxph(formula = surv_object ~ ctDNA.surveillance, data = circ_data)

n= 22, number of events= 12

              coef exp(coef) se(coef)      z Pr(>|z|)
ctDNA.surveillancePOSITIVE 1.9323    6.9055    0.6727 2.873  0.00407 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
ctDNA.surveillancePOSITIVE    6.905    0.1448    1.848    25.81

Concordance= 0.716 (se = 0.067 )
Likelihood ratio test= 9.95 on 1 df,  p=0.002
Wald test               = 8.25 on 1 df,  p=0.004
Score (logrank) test = 10.97 on 1 df,  p=9e-04
```

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 6.91 (1.85-25.81); p = 0.004"
```

#EFS by STMs at post-ACT - Stage II/III

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("CLIA Testicular_GL.csv")
circ_data <- circ_data[circ_data$ctDNA.available==TRUE,]
circ_data <- circ_data[circ_data$ACT==TRUE,]
circ_data <- circ_data[circ_data$Stage!="I",]
circ_data <- circ_data[circ_data$Markers.Surveillance!="",]
circ_datadf <- as.data.frame(circ_data)

survfit(Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)~Markers.Surveillance, data = circ_data)
```

```
Call: survfit(formula = Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event) ~
  Markers.Surveillance, data = circ_data)
```

	n	events	median	0.95LCL	0.95UCL
Markers.Surveillance=NEGATIVE	23	11	65.3	7.69	NA
Markers.Surveillance=POSITIVE	10	8	20.0	5.59	NA

Hide

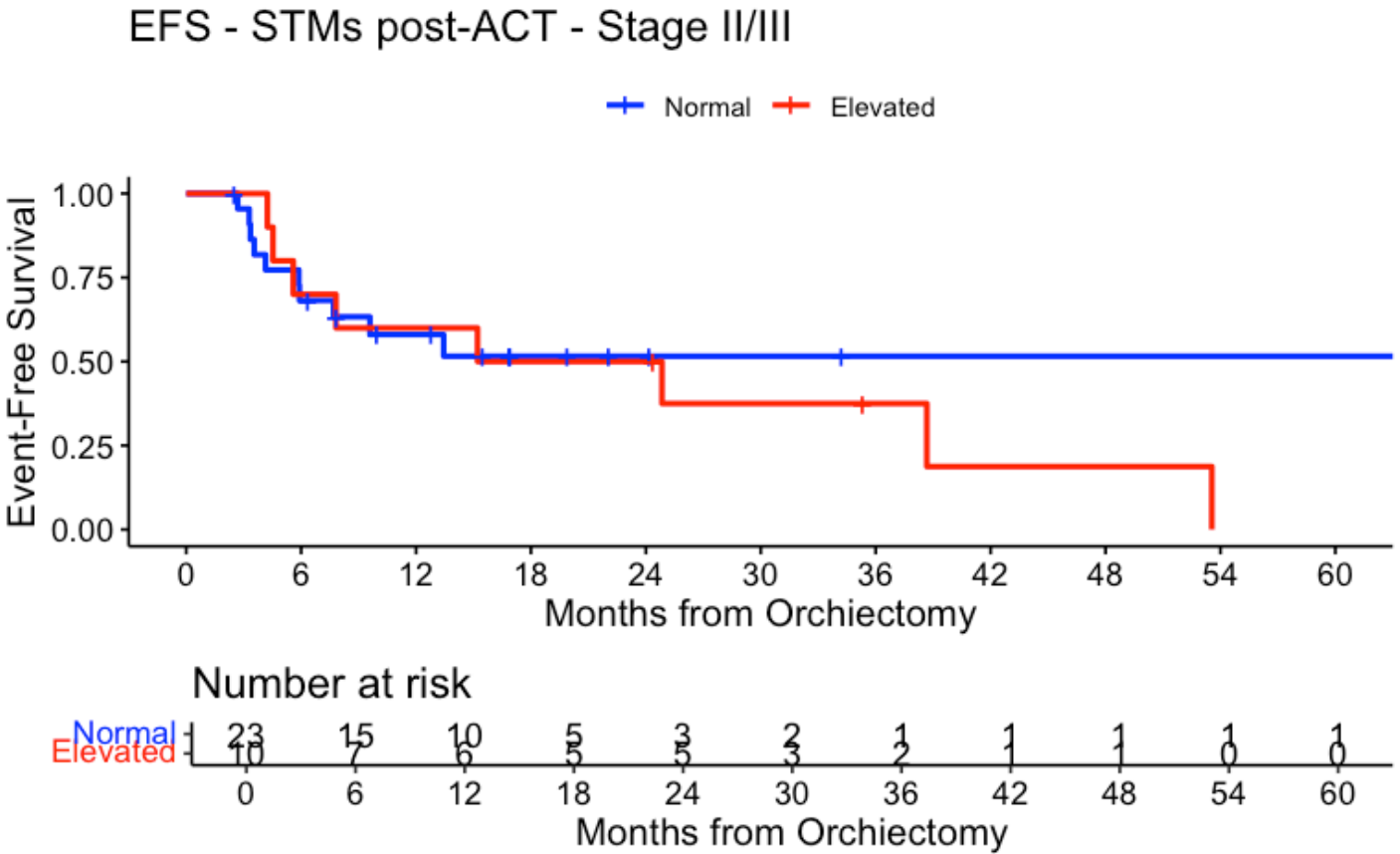
```
event_summary <- circ_data %>%
  group_by(Markers.Surveillance) %>%
  summarise(
    Total = n(),
    Events = sum(PFS.Event),
    Fraction = Events / n(),
    Percentage = (Events / n()) * 100
  )
print(event_summary)
```

Markers.Surveillance	Total	Events	Fraction	Percentage
<chr>	<int>	<int>	<dbl>	<dbl>
NEGATIVE	23	11	0.4782609	47.82609
POSITIVE	10	8	0.8000000	80.00000

2 rows

Hide

```
surv_object <-Surv(time = circ_data$PFS.months, event = circ_data$PFS.Event)
KM_curve <- survfit(surv_object ~ Markers.Surveillance, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - STMs post-ACT - Stage II/III", ylab= "Event-Free Survival", xlab="Months from Orchiectomy", legend.labs=c("Normal", "Elevated"), legend.title="")
```

Hide

```
summary(KM_curve, times= c(12, 24))
```

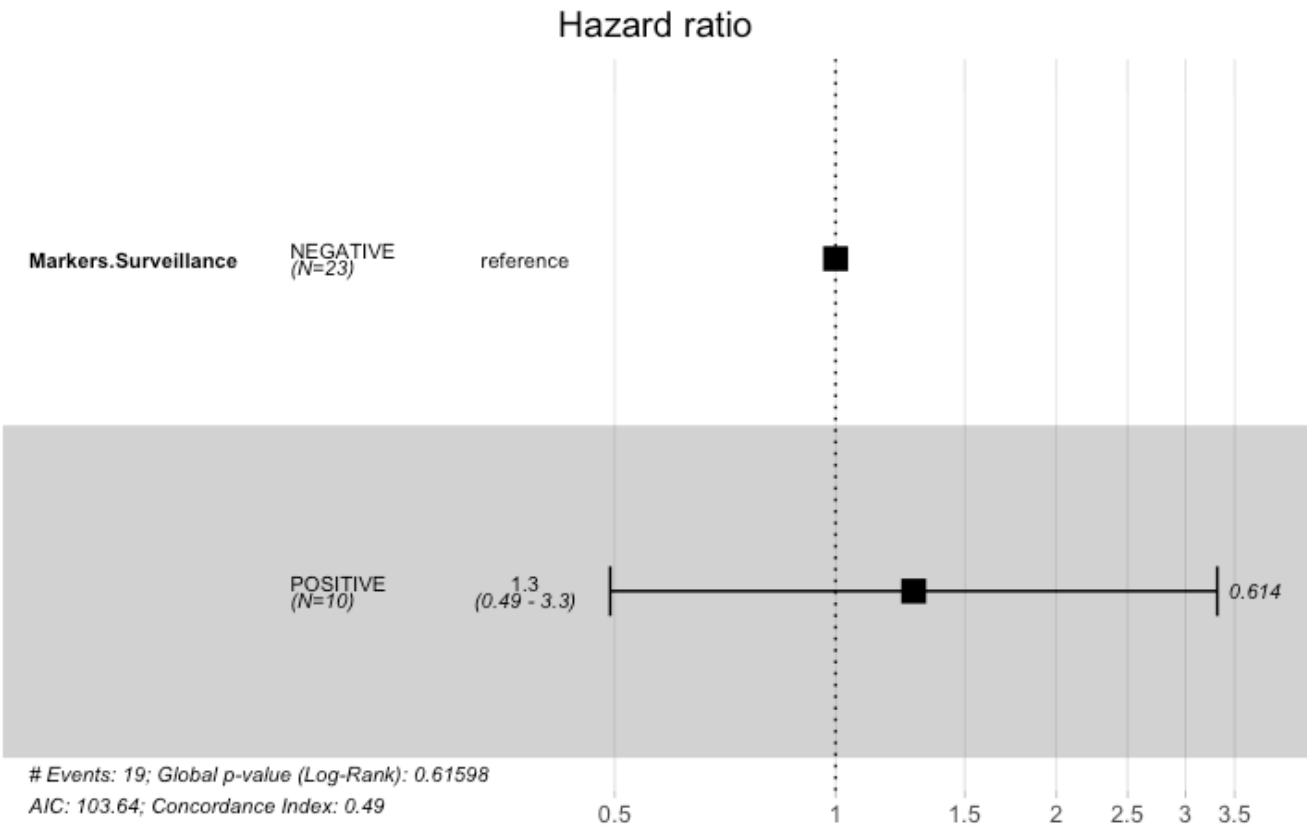
```
Call: survfit(formula = surv_object ~ Markers.Surveillance, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

Markers.Surveillance=NEGATIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
12	10	9	0.580	0.107	0.346	0.756	
24	3	1	0.516	0.113	0.281	0.708	

Markers.Surveillance=POSITIVE							
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI	
12	6	4	0.6	0.155	0.253	0.827	
24	5	1	0.5	0.158	0.184	0.753	

Hide

```
circ_data$Markers.Surveillance <- factor(circ_data$Markers.Surveillance, levels=c("NEGATIVE", "POSITIVE"))
cox_fit <- coxph(surv_object ~ Markers.Surveillance, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ Markers.Surveillance, data = circ_data)
```

n= 33, number of events= 19

	coef	exp(coef)	se(coef)	z	Pr(> z)
Markers.SurveillancePOSITIVE	0.2453	1.2780	0.4862	0.505	0.614

	exp(coef)	exp(-coef)	lower .95	upper .95
Markers.SurveillancePOSITIVE	1.278	0.7825	0.4928	3.314

Concordance= 0.49 (se = 0.058)

Likelihood ratio test= 0.25 on 1 df, p=0.6

Wald test = 0.25 on 1 df, p=0.6

Score (logrank) test = 0.26 on 1 df, p=0.6

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 1.28 (0.49-3.31); p = 0.614"
```